

Color Versions of Figures

This file provides the color versions of all the figures, and is intended as a **supplement to the printed** version of the book. The electronic versions contain the color versions by default.

For the sake of consistency, all figures that are marked with the comment “Color version available at www.wiley.com/go/molisch/wireless3e” are contained here, even when they are easily read in greyscale. However, the main importance of this file is in providing the figures that would be difficult to read / discern different curves in the printed greyscale reproduction.

Chapter I

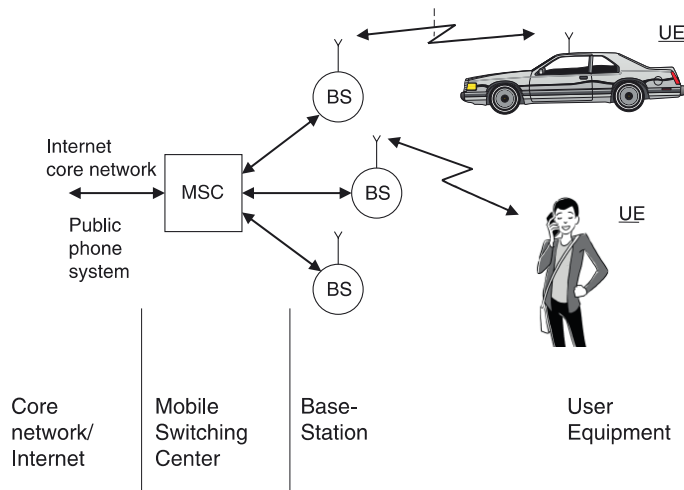


Figure 1.1 Principle of a cellular system. In this figure: MSC, Mobile Switching Center; BS, Base Station; UE, User Equipment.

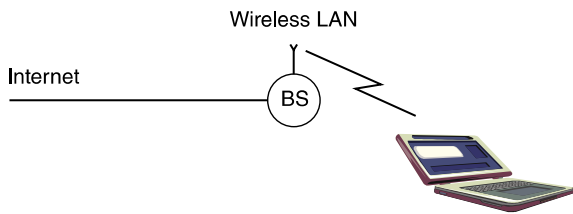


Figure 1.2 Principle of a simple WLAN.

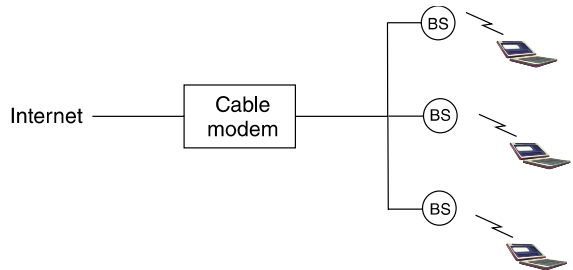


Figure 1.3 Principle of an enterprise WLAN.

Chapter 2

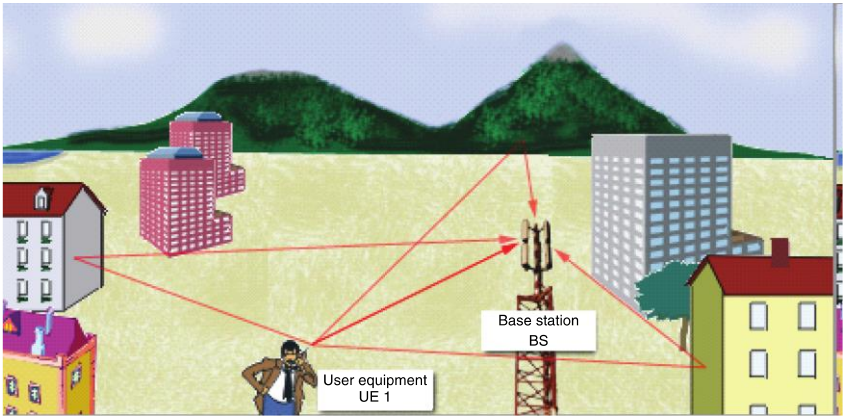


Figure 2.1 Multi-path propagation.

Chapter 3

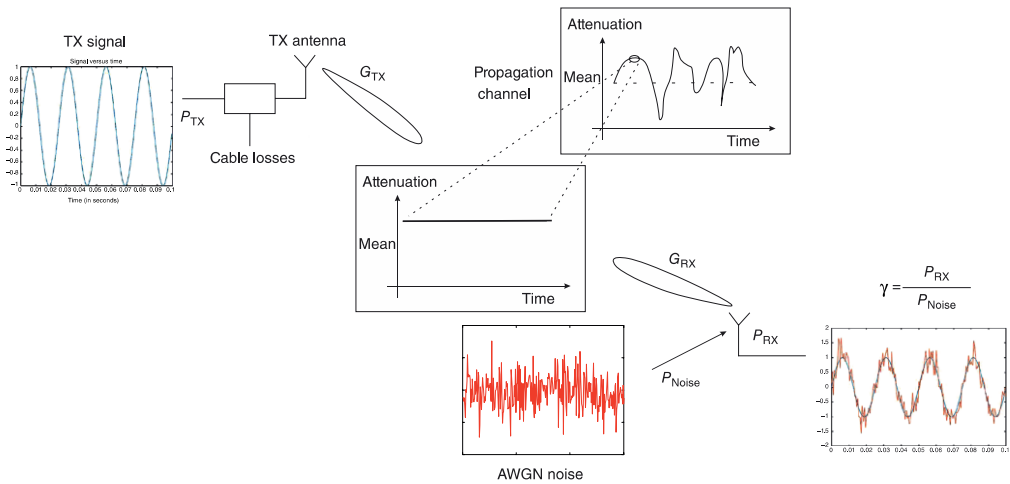
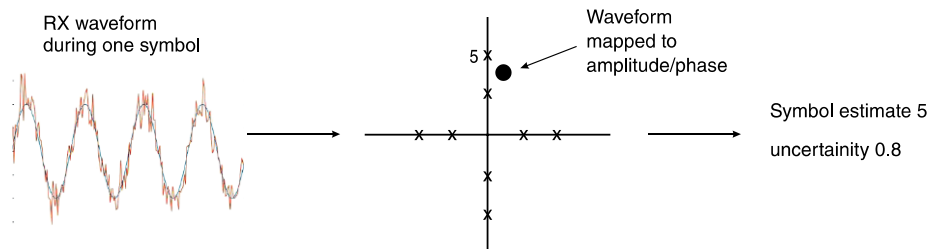
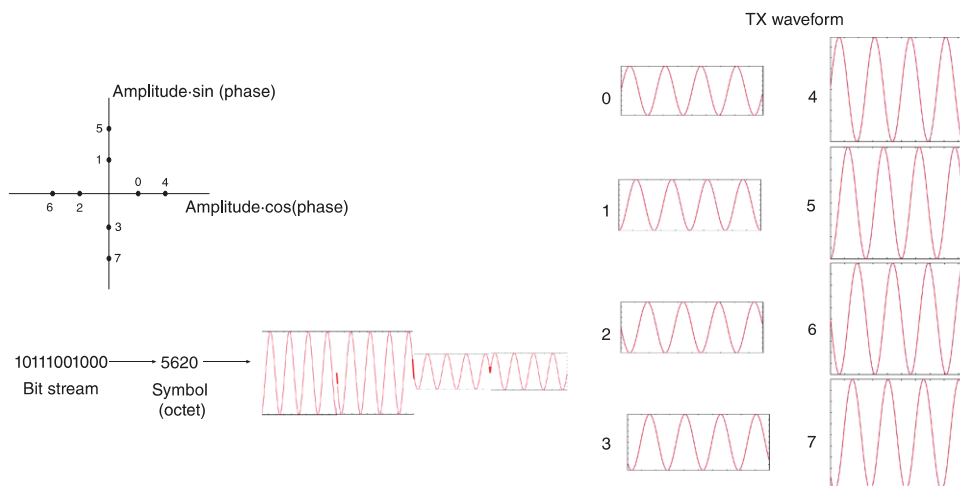
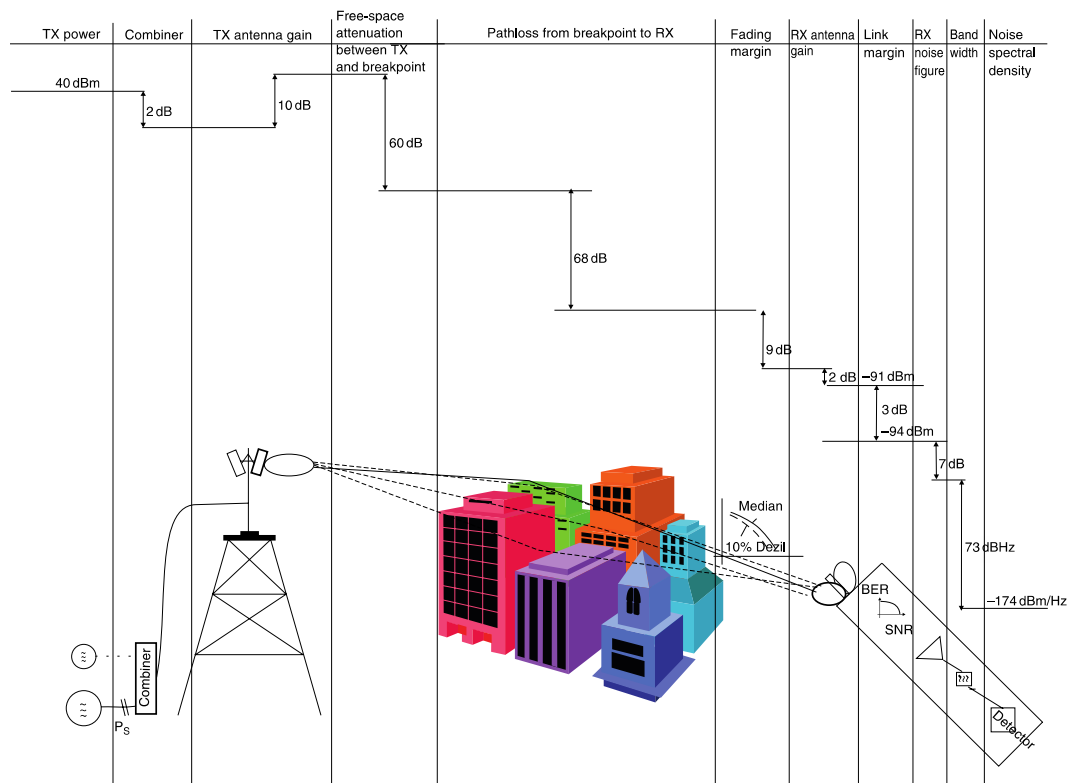


Figure 3.1 Block diagram of a wireless system with flat fading. Note that the channel changes slowly, so that at a given time (where signal and noise vary) it is effectively constant.



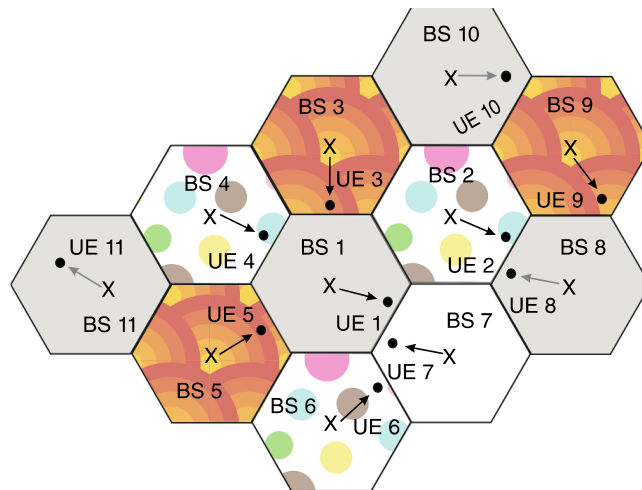


Figure 3.21 Base stations in different cells using different frequencies, indicated by shading.

Chapter 4

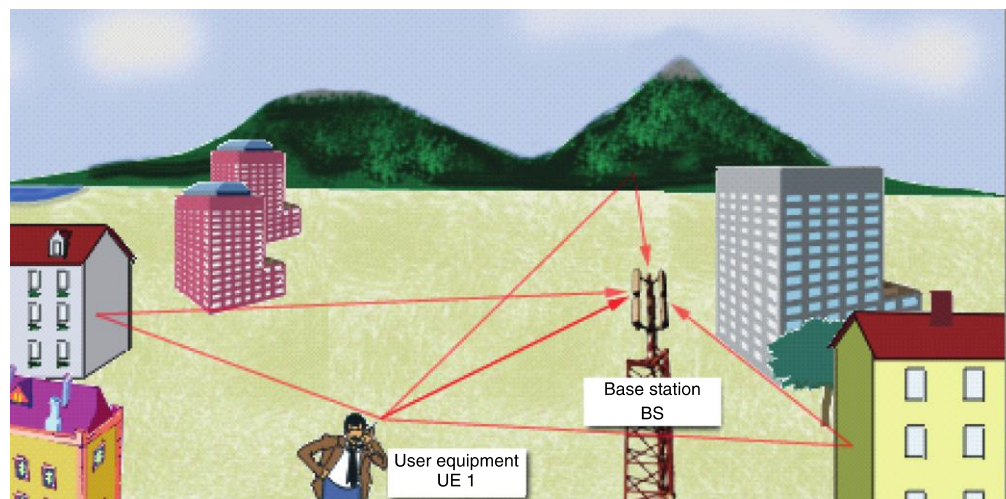


Figure 4.1 Multi-path propagation.

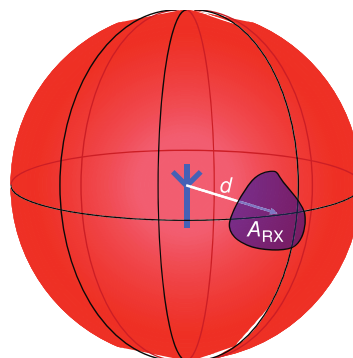


Figure 4.2 Received power area density at distance d , and collection of power by antenna area A_{RX} .

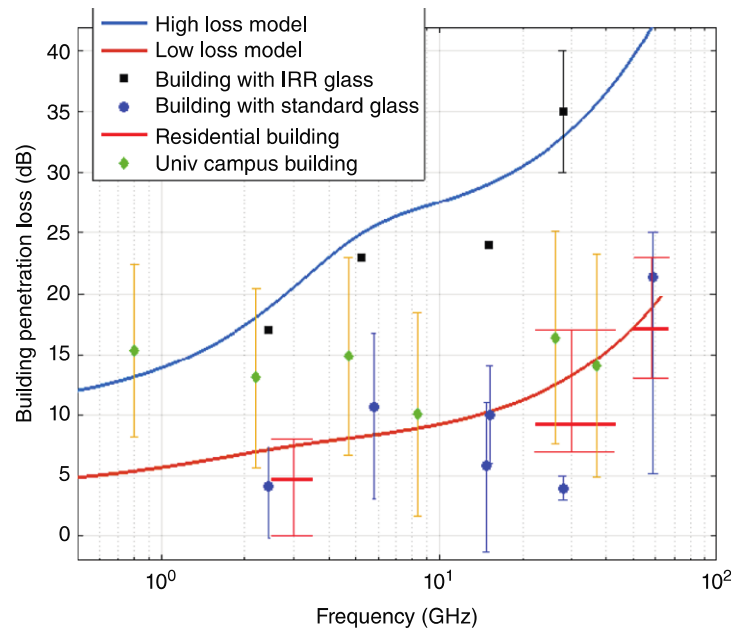


Figure 4.5 Building penetration as function of frequency.
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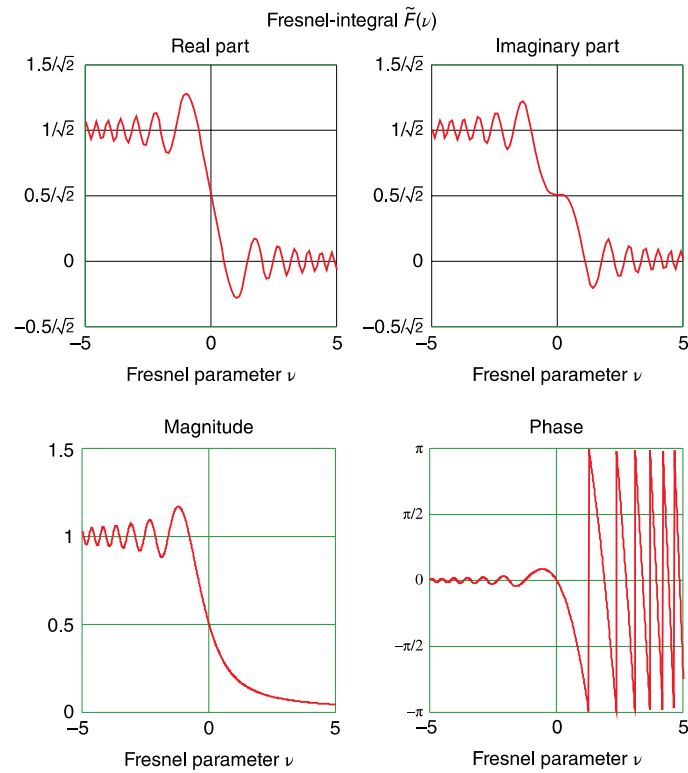


Figure 4.8 Fresnel integral.

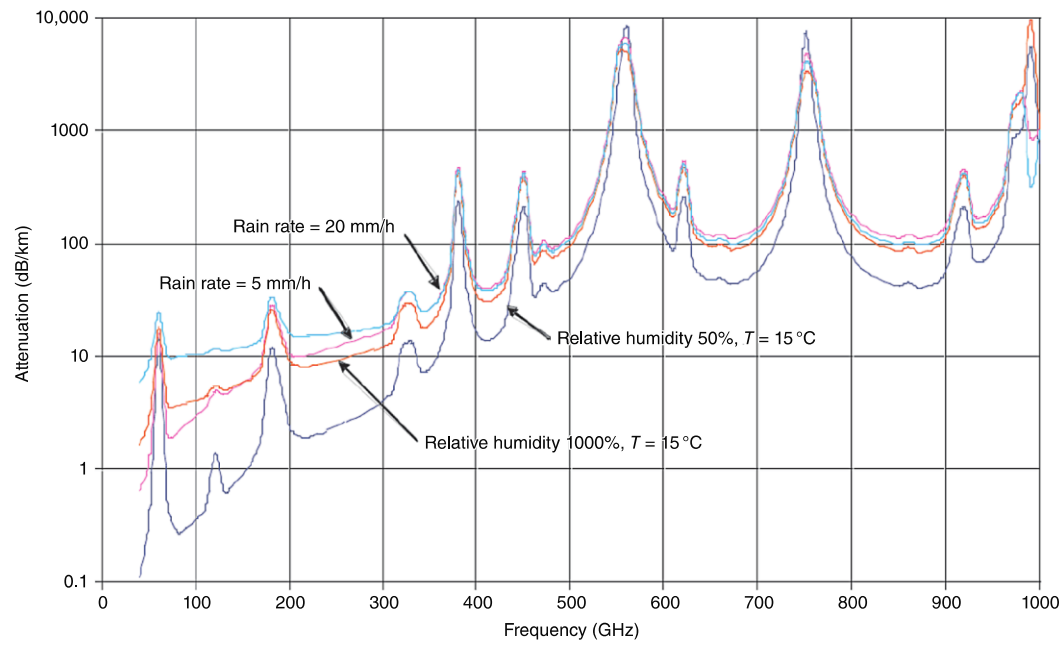


Figure 4.18 Atmospheric attenuation 40–1000 GHz.
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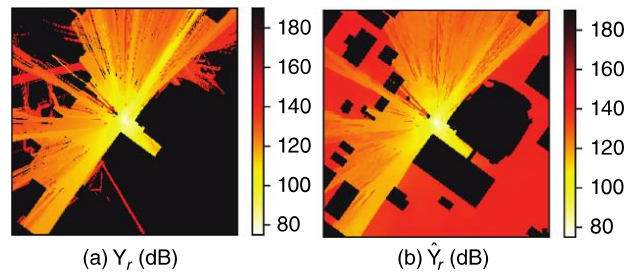


Figure 4.22 Comparison between ray tracing (a) and ML prediction (b) in a microcell in Houston.
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Chapter 7

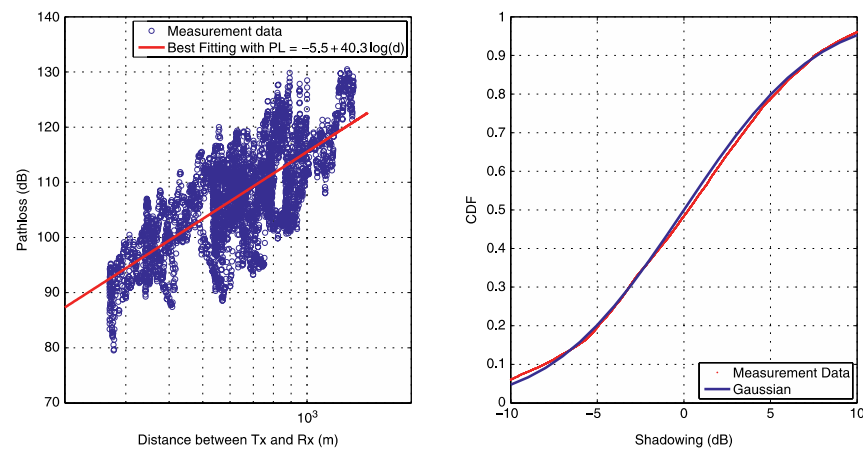


Figure 7.2 Example for fitting of pathloss and shadowing from measurement data in suburban environment.

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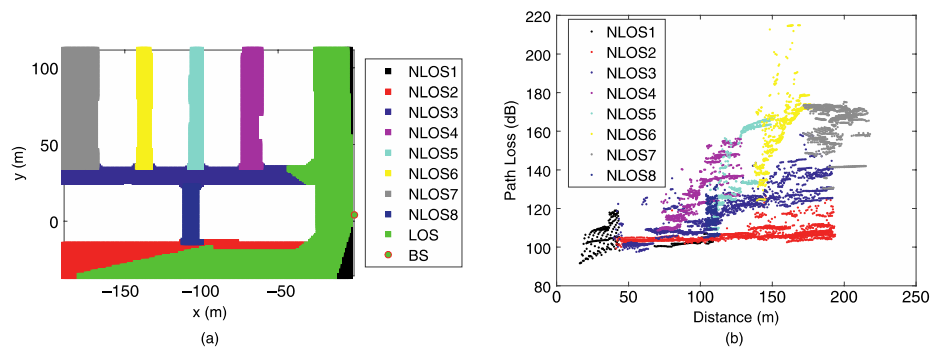


Figure 7.4 (a) Environment map with color coding of streets. (b) Pathloss as a function of link distance in NLOS, different greyscale shades indicate different streets.

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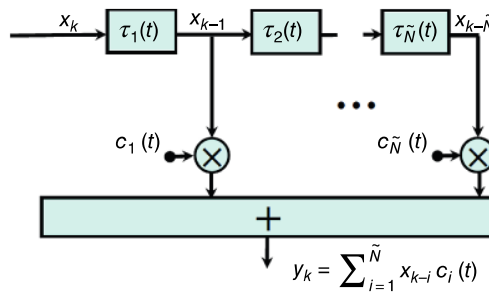


Figure 7.5 Tapped-delay line model.

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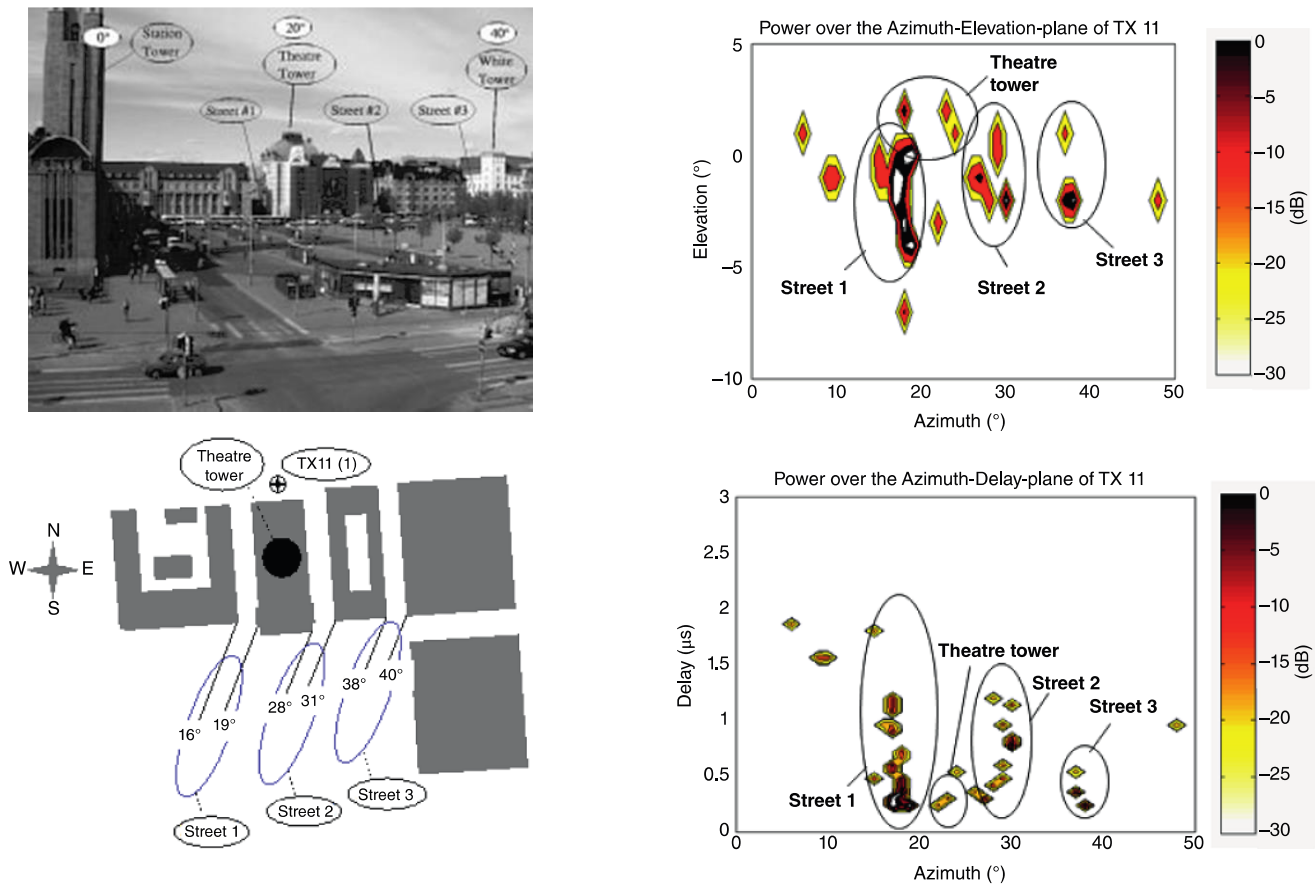


Figure 7.8 Photograph from BS location, and APS with propagation paths for the clusters. Theatre tower is a tall building acting as far IO.

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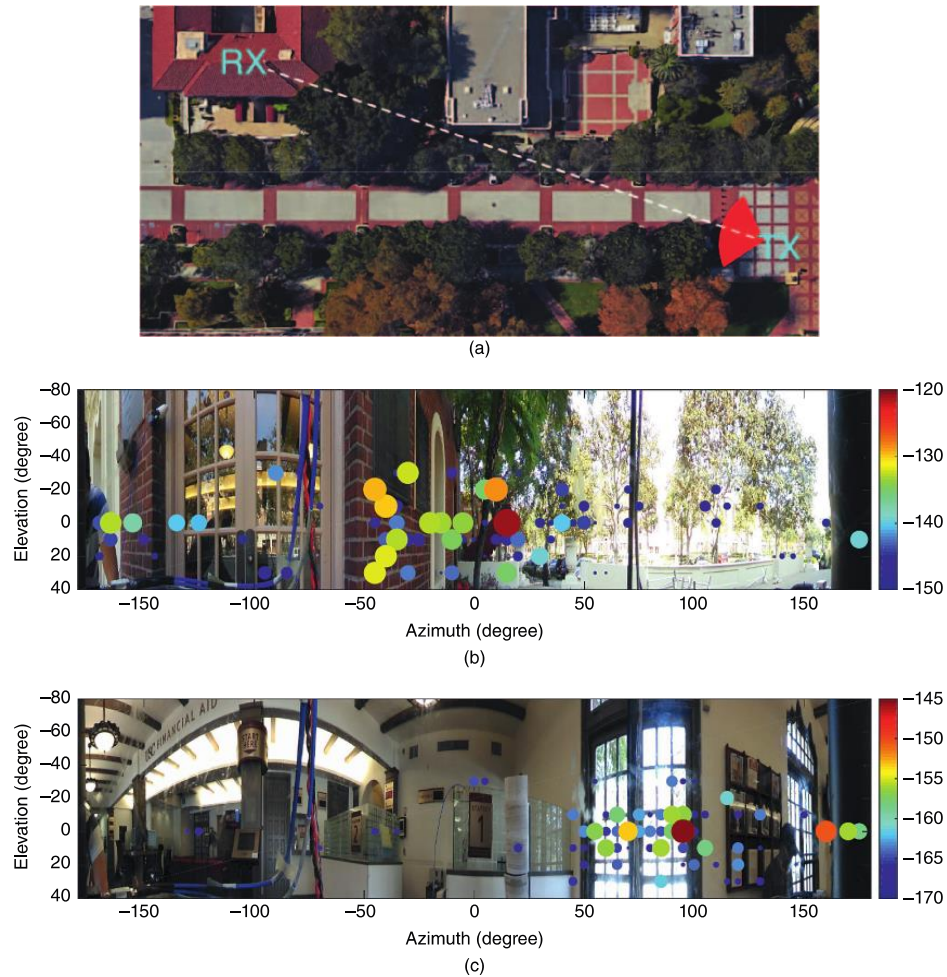


Figure 7.9 Bird's eye view of propagation scenario (a). Strengths and directions of mm-wave MPCs superimposed on photograph from UE location outside (b) and inside (c) a building. It can be seen that components that are strong outdoors, but incident on the building at a grazing angle, vanish or show very weak amplitude indoors, while components entering through the glass door show significant strength.

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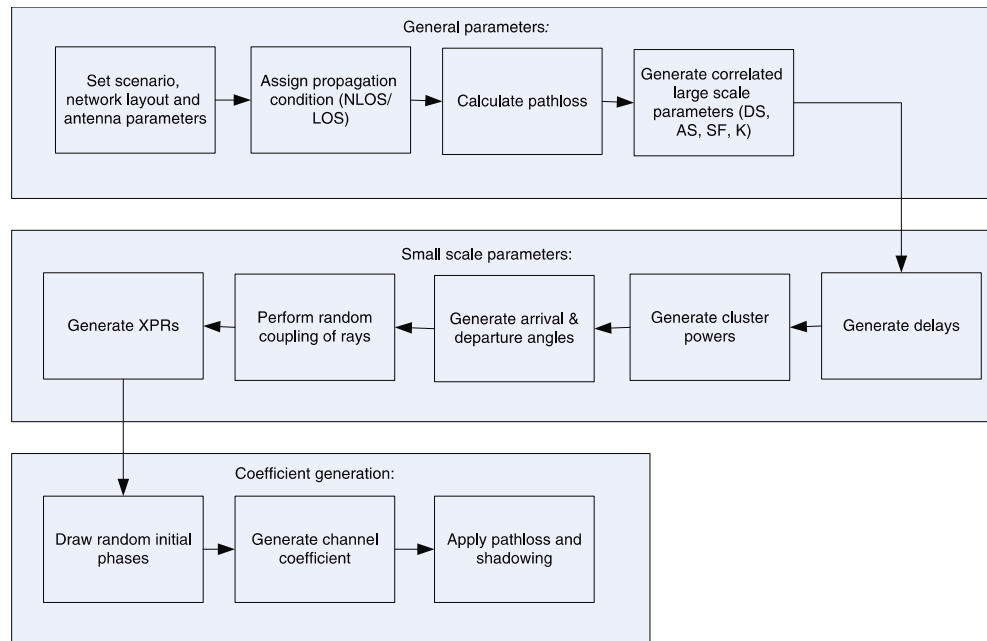


Figure 7.10 Standardized 3GPP / ITU-R channel impulse response generation procedure for propagation channel models in 4G and 5G systems. Reproduced with permission from [3GPP TR 38.901] © 2019. 3GPP™ TSSs and TRs are the property of ARIS, ATIS, CCSA, ETSI, TTA and TTC, who jointly own the copyright in them. They are subject to further modifications and are therefore provided, to you “as is” for information purposes only. Further use is strictly prohibited.

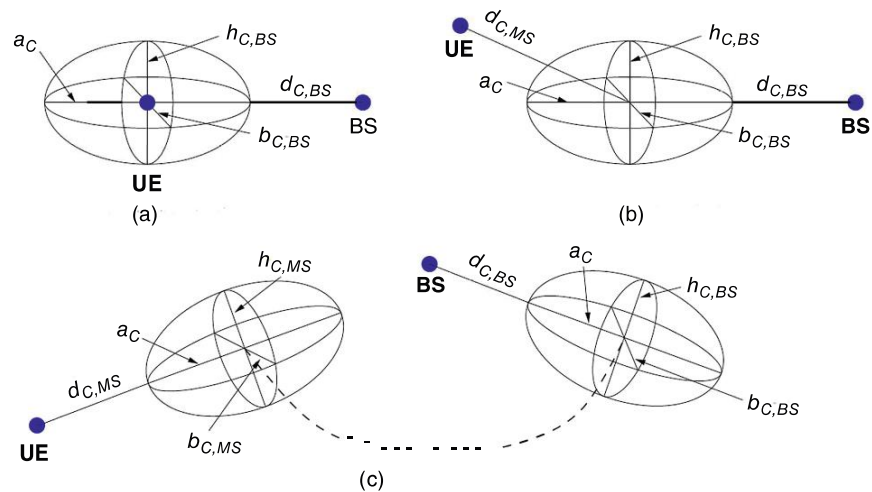


Figure 7.13 Three kinds of clusters: (a) local cluster, (b) far single cluster, (c) twin cluster. The acronym MS in the figure is UE in the notation of this book. Reproduced with permission from [Verdone and Zanella 2012] © Springer.

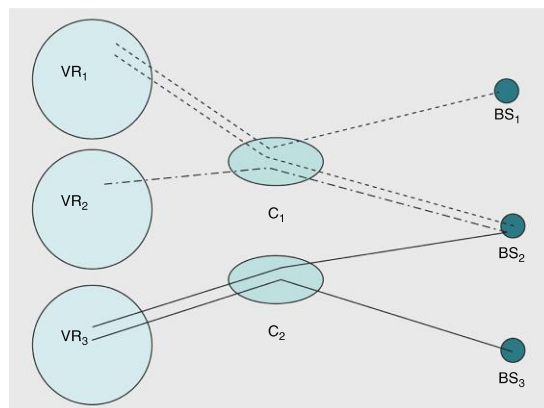


Figure 7.15 Examples of common clusters. The clusters C_1 and C_2 are associated to three BSs and three users in the visibility regions. The lines define the visibility of the different users to the BSs. Reproduced with permission from [Liu et al. 2012] © IEEE.

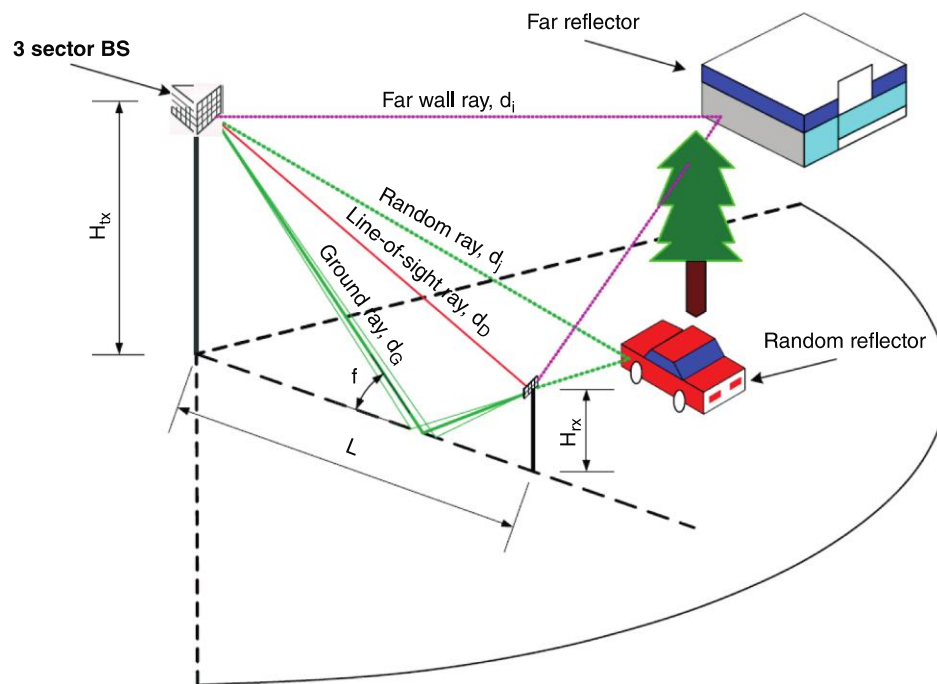


Figure 7.16 Components of a semi-deterministic channel model. Reproduced with permission from [Weiler et al. 2016] © EuraSIP.

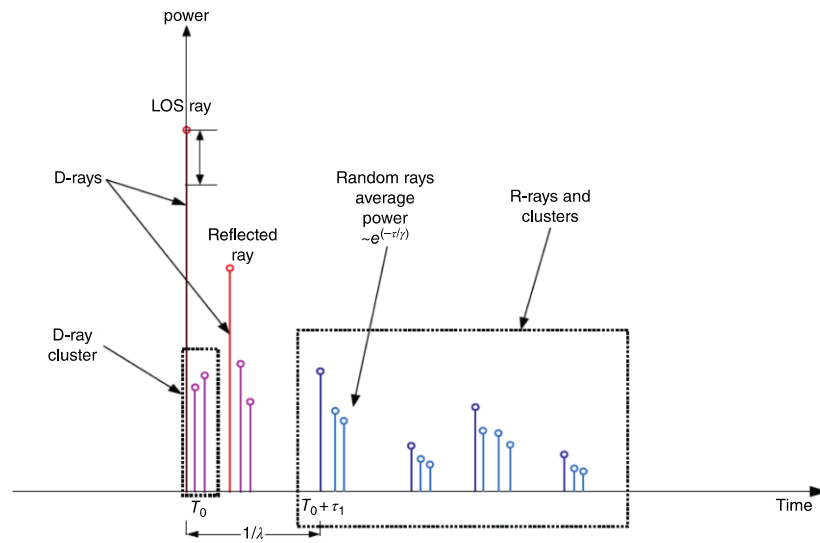


Figure 7.17 Structure of a semi-deterministic channel model.
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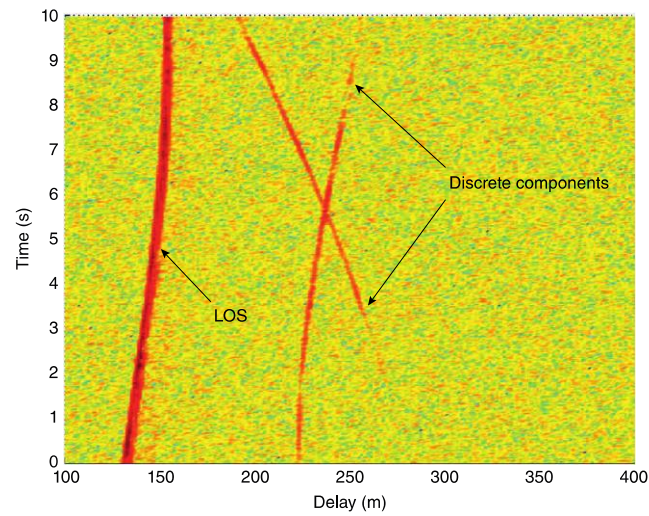


Figure 7.18 Example plot of a time-varying power delay profile of V2V communications on a highway. Delay is multiplied with speed of light.

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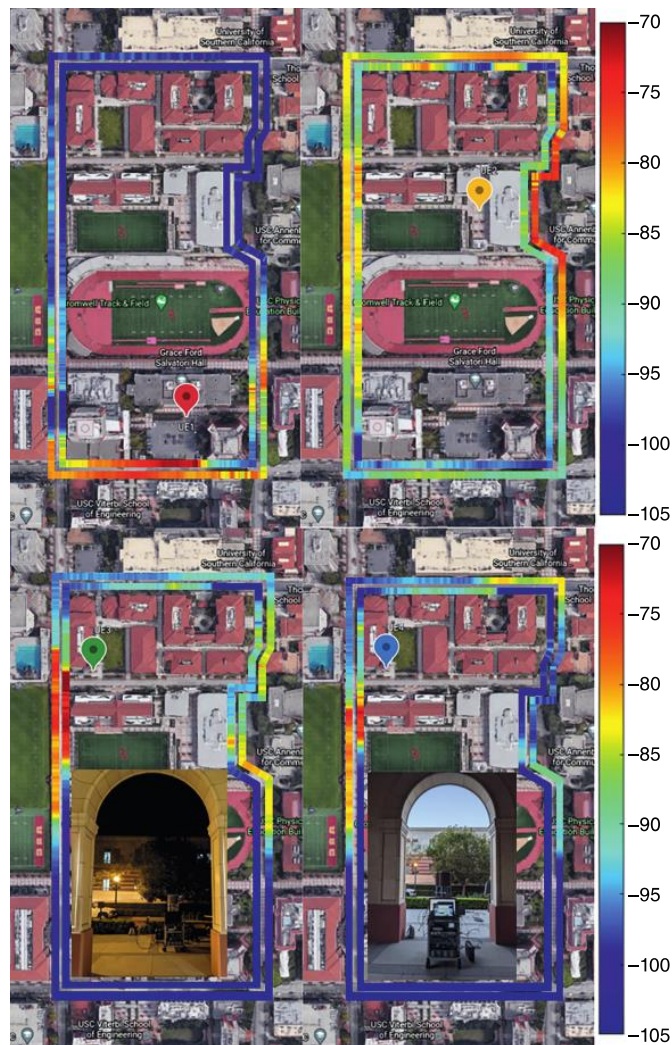


Figure 7.19 Channel measurement environment in an urban environment (university campus): colorbars show channel gains (in dB) between all AP locations (drone positions) along the trajectory at 35m (inner loop)/70m height (outer loop) and four different UE locations.

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Chapter 8

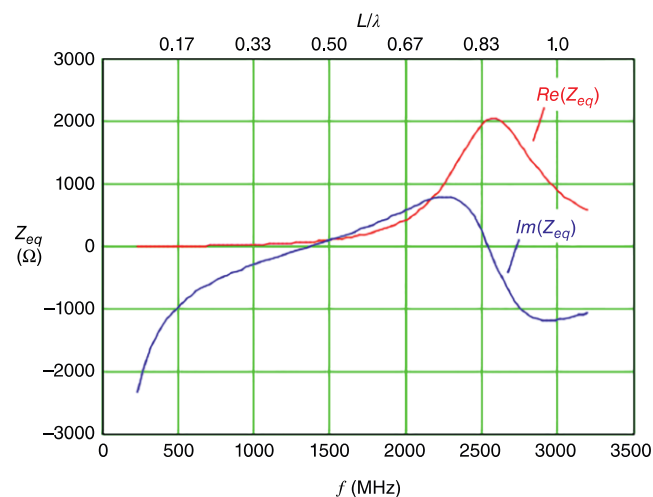


Figure 8.4 Input impedance for a thin dipole ($L_a/a = 100$), $L_a = 10$ cm.

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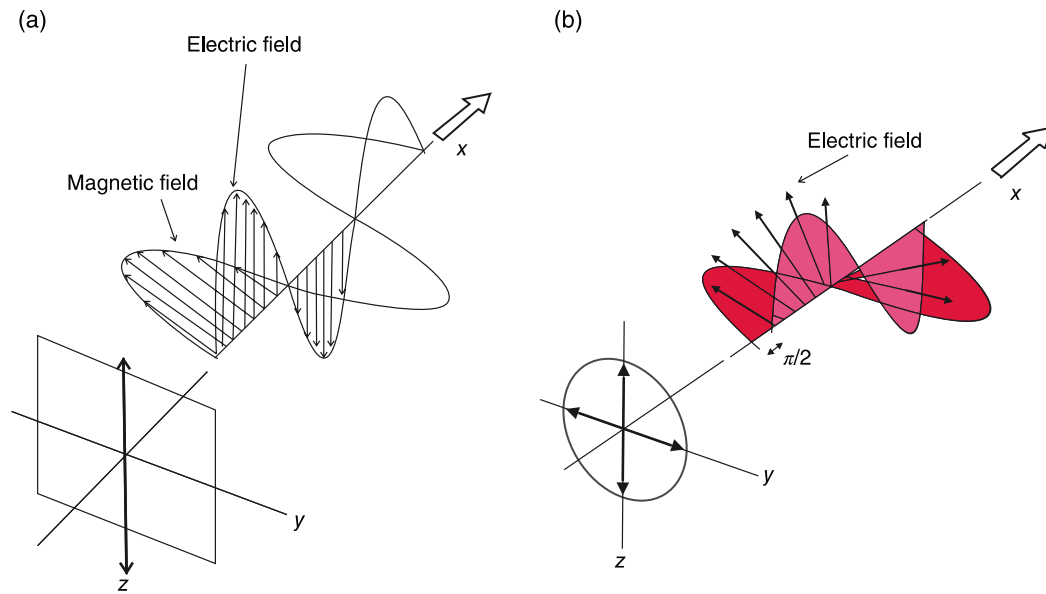


Figure 8.6 Polarization vectors: (a) Linear polarization. (b) Circular polarization.

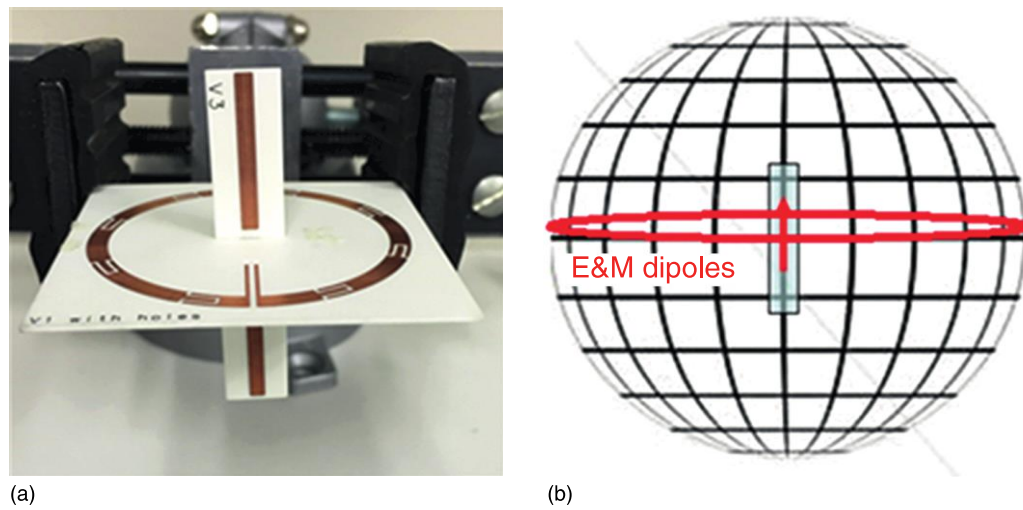


Figure 8.8 Electric dipole and loop antenna (magnetic dipole) for dual-polarized collocated antennas. Reproduced with permission from [Mirza et al. 2017] © Intech Open.

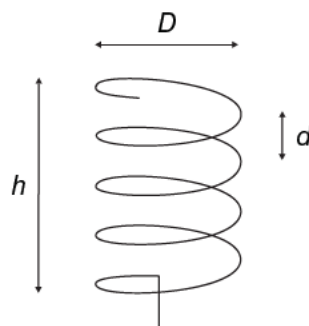


Figure 8.10 Geometry of a helical antenna.

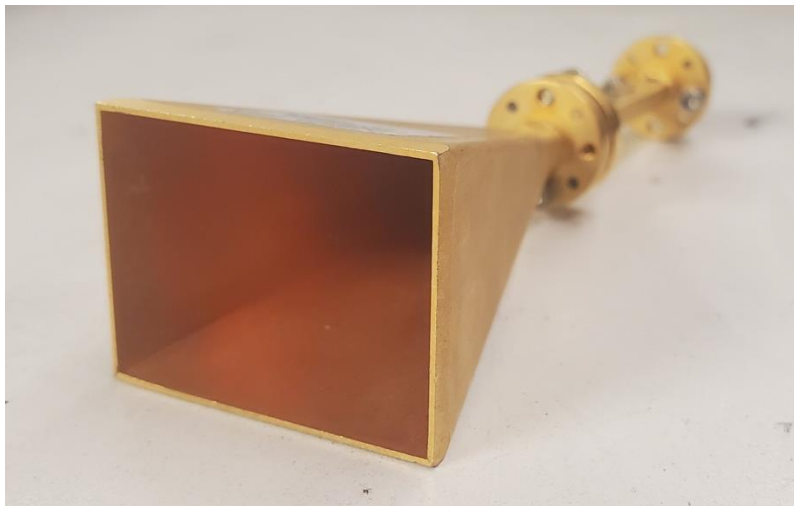


Figure 8.16 60 GHz standard gain horn antenna.



Figure 8.18 Picture of parabolic antenna (bottom) and Yagi–Uda antenna (top).
Unknown Author © Wikimedia Commons.

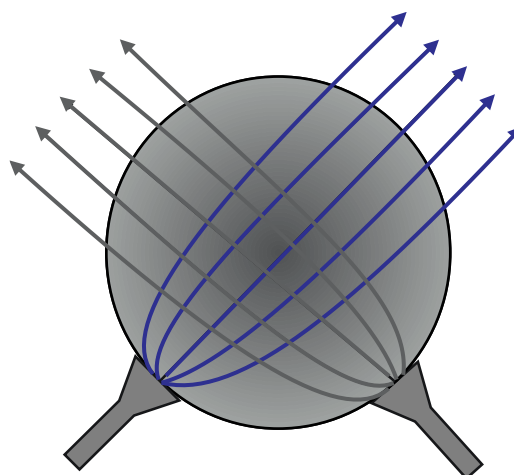


Figure 8.20 Luneberg lens antenna. Darker shading indicates higher dielectric constant.

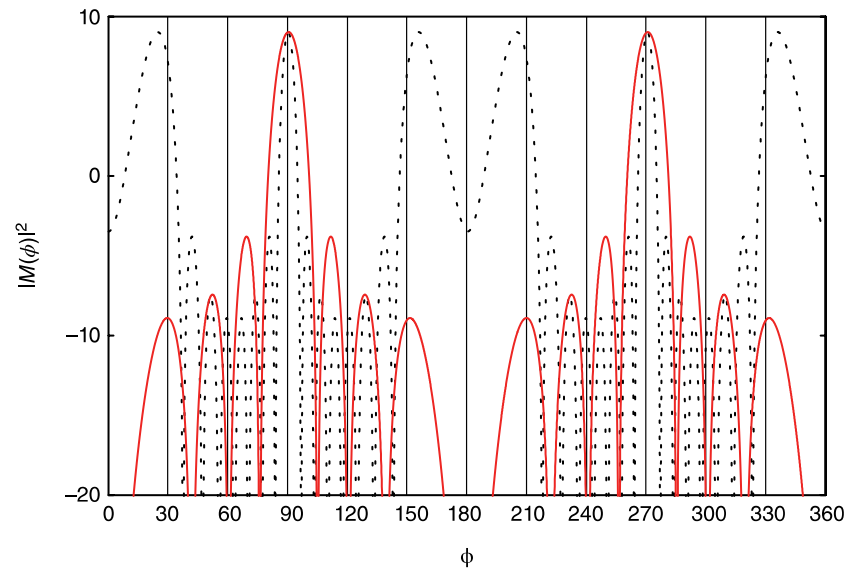


Figure 8.22 Array factor $M(\phi)$ of a uniform linear array with $d = 0.5\lambda$ (solid) and $d = 1.1\lambda$ (dotted), and $N_r = 8$ antenna elements.

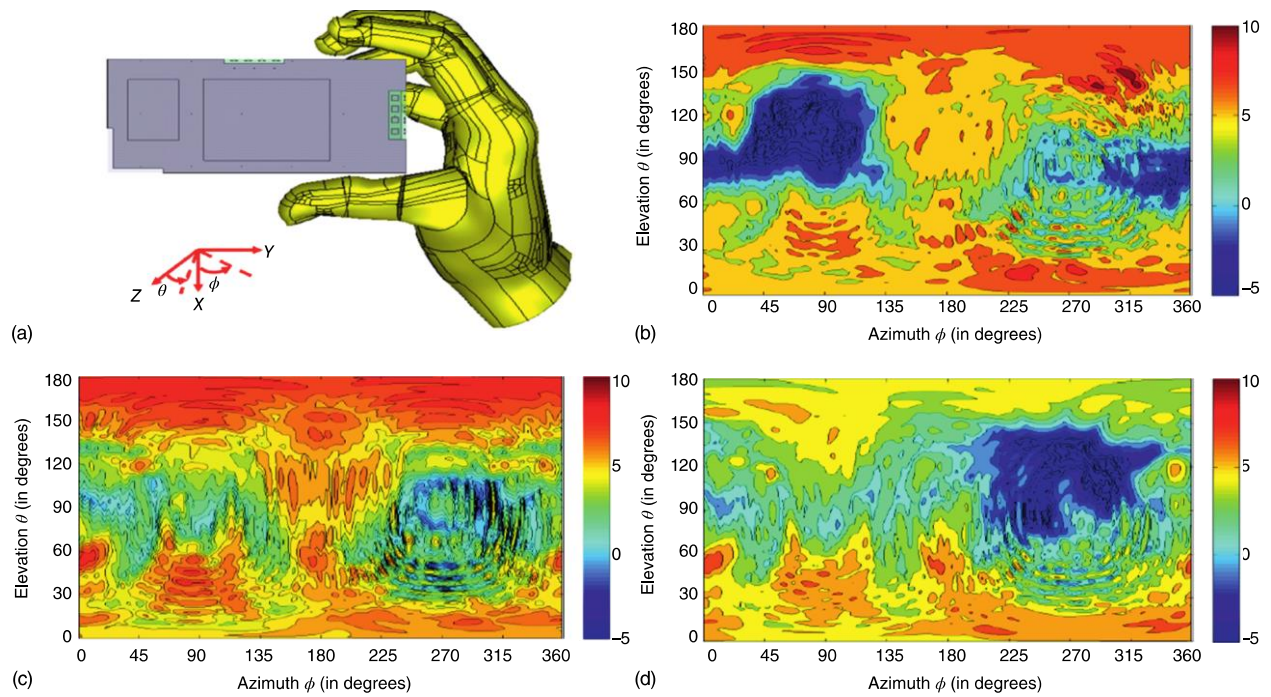


Figure 8.26 (a) A typical UE design with multiple subarrays (top and long edges) in Landscape mode. Received gain as a function of azimuth and elevation angles for the UE design at 28 GHz in (b) freespace mode, and with hand blocking in (c) landscape and (d) portrait modes. Reproduced with permission from [Raghavan et al. 2018] © IEEE.

Chapter 13

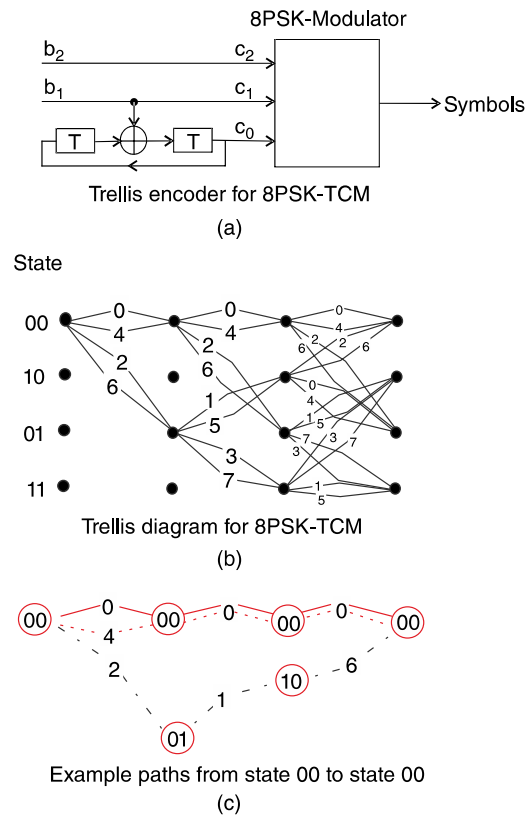


Figure 13.11 Encoder and trellis diagram for a rate-2/3 8-PSK trellis-coded modulation with four trellis states. (a) Trellis-coded modulator structure, (b) trellis diagram, (c) example paths.
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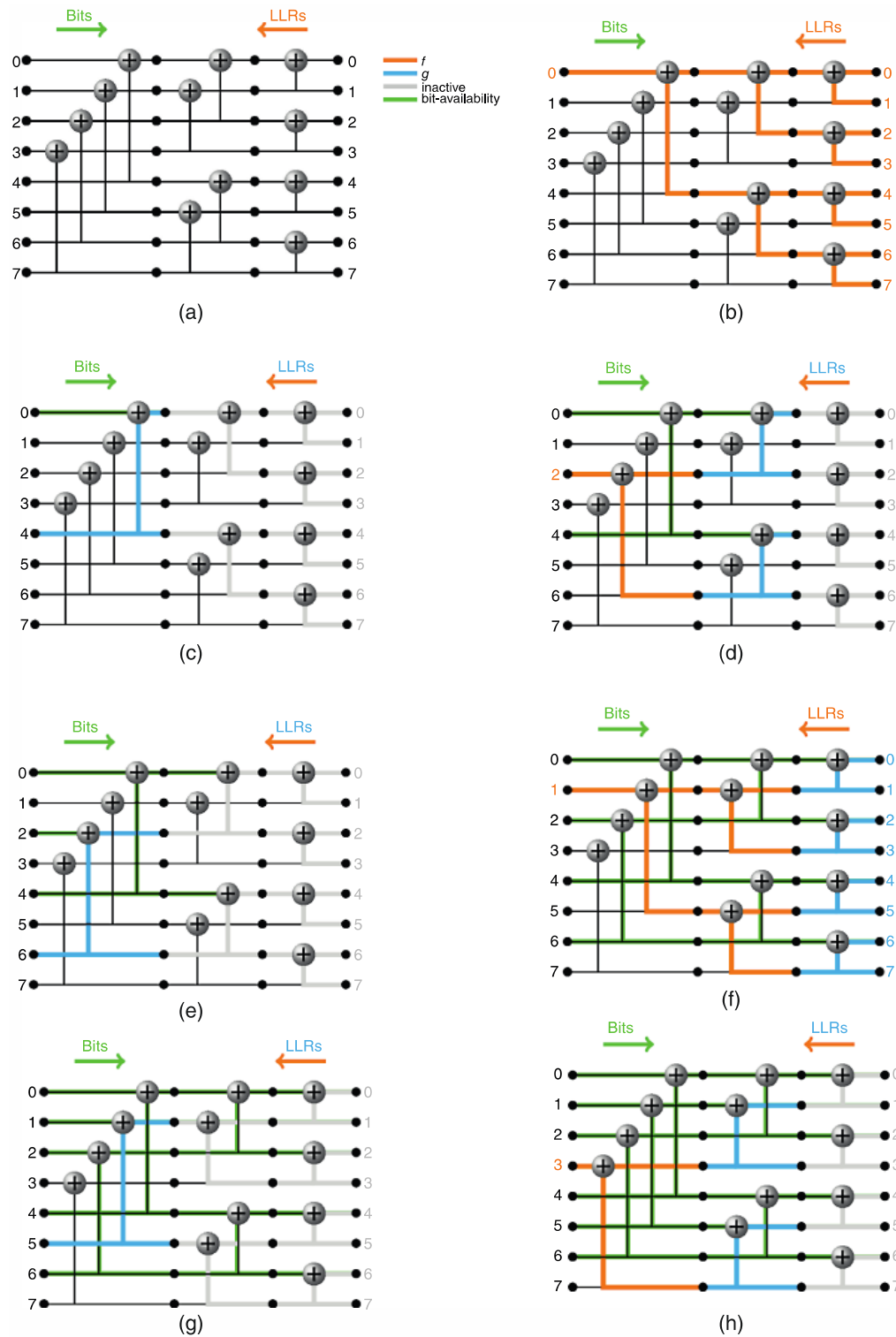


Figure 13.28 Decoding of a polar code by successive cancellation decoding; see text for the description of the steps (a)–(h).
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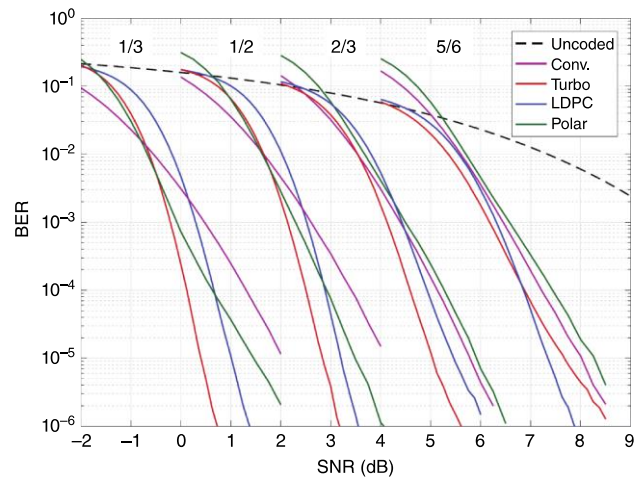


Figure 13.29 Comparison of convolutional, turbo, LDPC, and polar codes at different code rates with source block rate $K = 256$ in AWGN with BPSK modulation.

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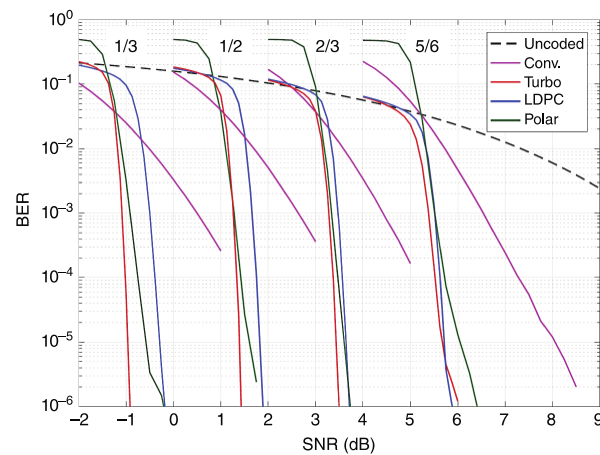


Figure 13.30 Comparison of convolutional, turbo, LDPC, and polar codes at different code rates with source block size $K = 8192$ in AWGN with BPSK modulation.

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Chapter 15

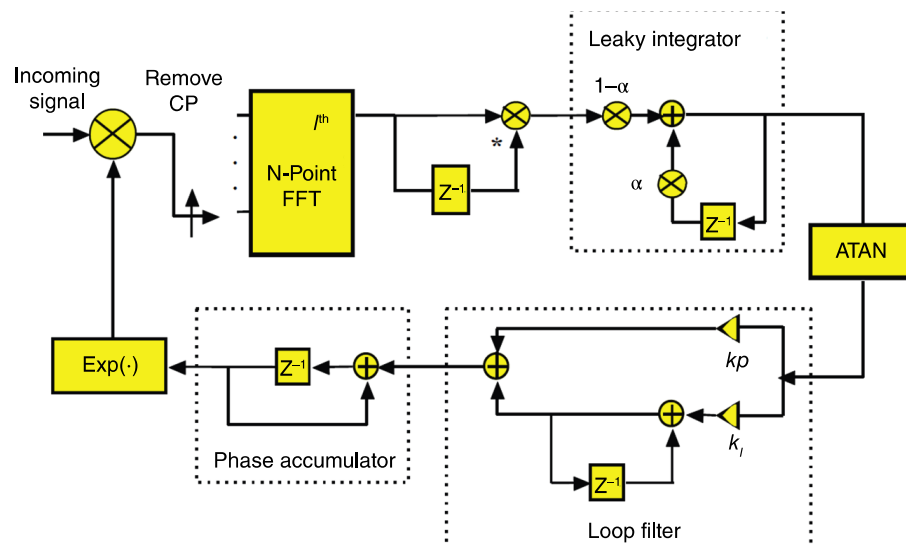


Figure 15.18 Receiver for tracking CFO with closed-loop structure.

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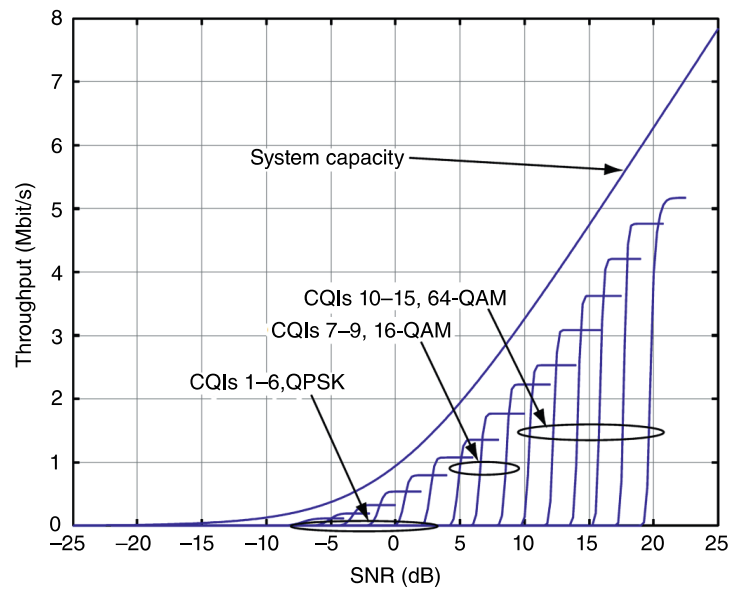


Figure 15.20 System capacity, and throughput achievable with different modulation formats and code rates (choice of which depends on the “channel quality indicator” CQI) in LTE.

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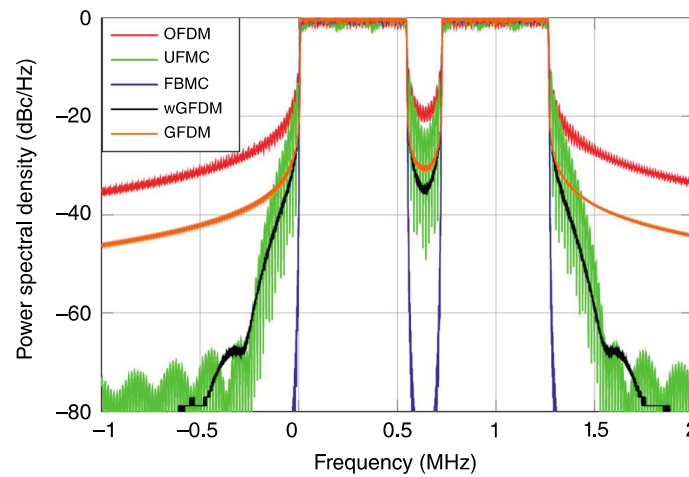


Figure 15.21 Power Spectral Density of different multi-carrier modulation methods.

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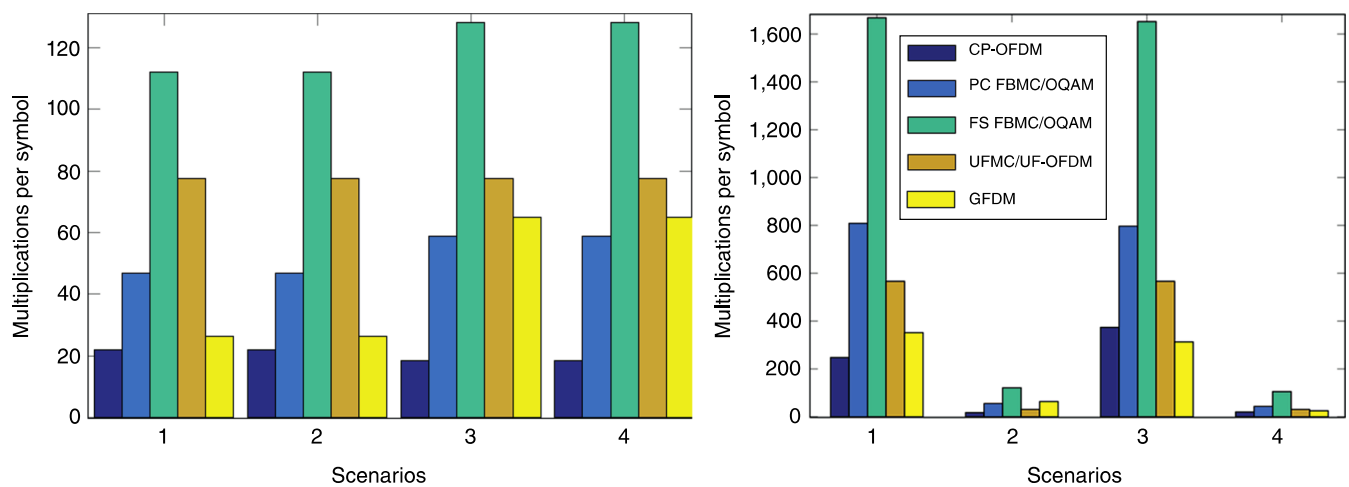


Figure 15.22 Complexity of different multi-carrier schemes in units of real multiplications at BS (left) and UE (right). Scenarios: (1) Downlink with multiple UEs frequency multiplexed, (2) Downlink single UE, (3) Uplink multiple UEs frequency multiplexed, (4) Uplink single UE.

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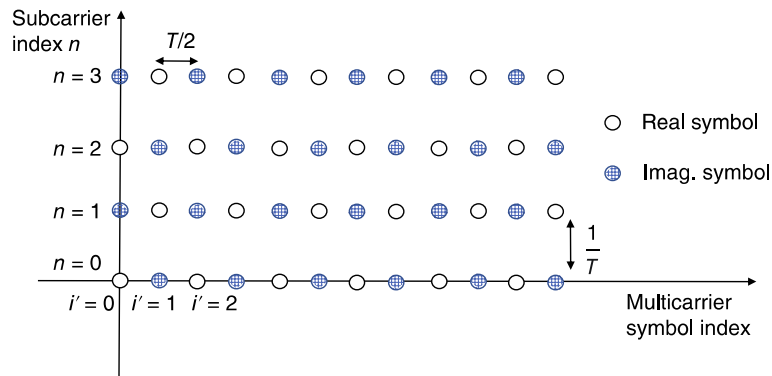


Figure 15.24 Staggered lattice for FBMC/OQAM.

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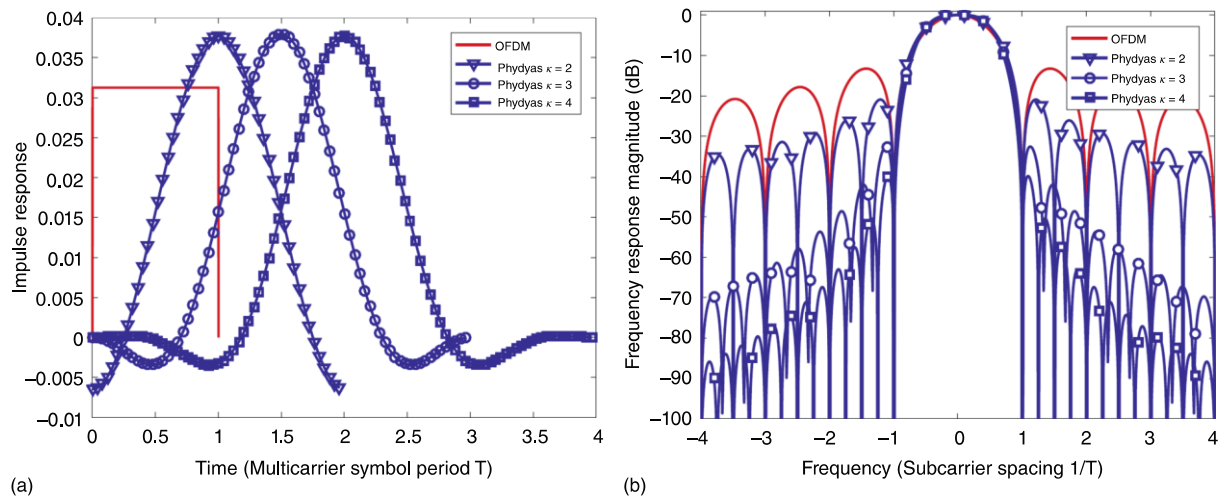


Figure 15.25 Pulses for FBMC/OQAM, compared to standard rectangular pulse. (a) Time domain representation and (b) frequency domain.

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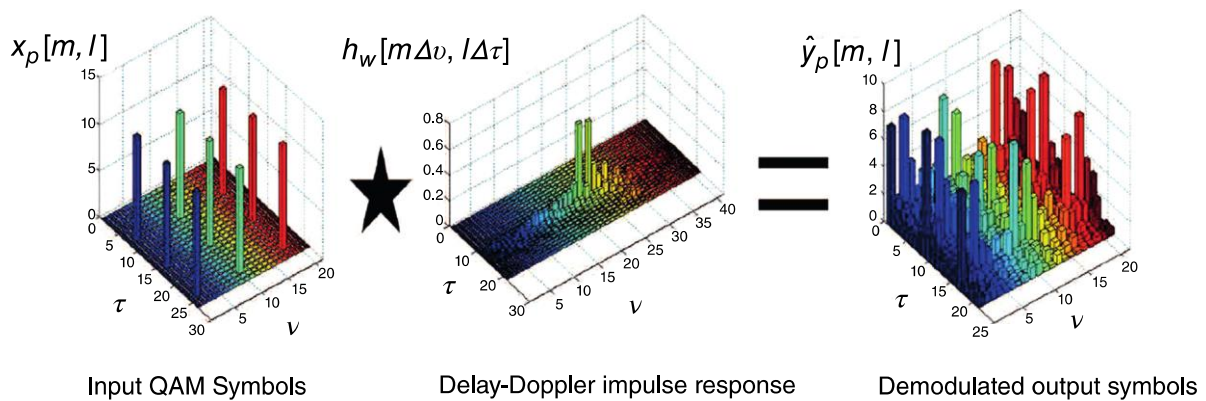


Figure 15.28 Response with modulation symbols in the delay/Doppler domain.

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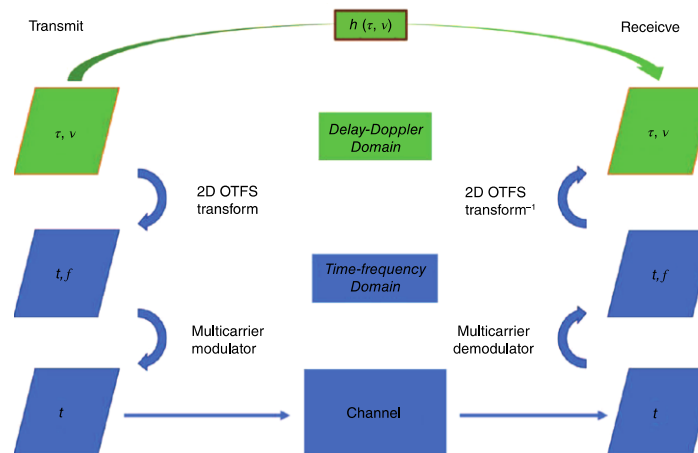


Figure 15.29 Signal flow in an implementation of OTFS as time-frequency overlay.

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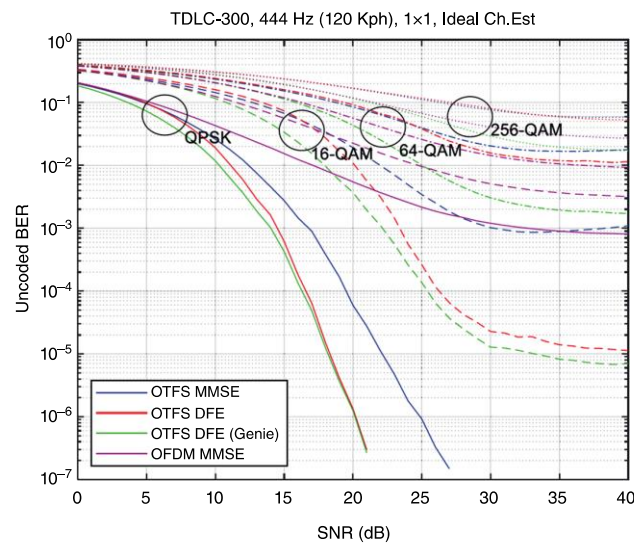


Figure 15.30 Unencoded BER for 4QAM/16QAM/64QAM/256QAM, TDL-C channel model, 120 kmph. Curves show OFDM with MMSE equalizer, and OTFS with (i) MMSE equalizer, (ii) DFE equalizer with error propagation, and (iii) DFE equalizer with genie feedback.

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Chapter 16

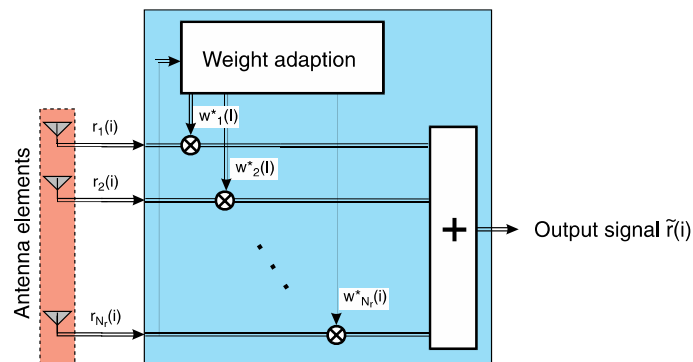


Figure 16.1 Linear combination of antenna signals.

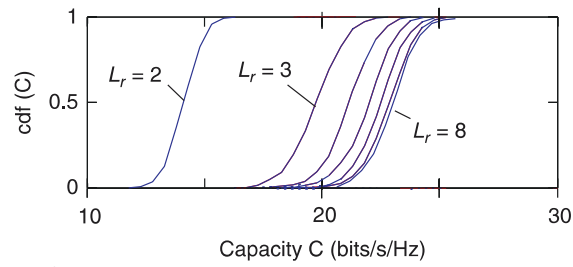


Figure 16.25 Capacity for a spatial multiplexing system with $N_r=8$, $N_t=3$, $\text{SNR}=20$ dB, and $L_r = 2, 3, \dots, 8$

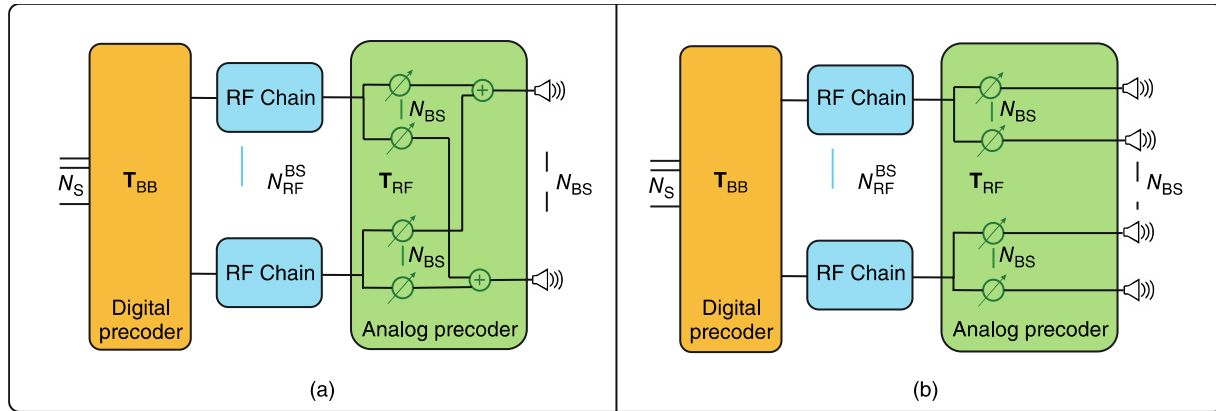


Figure 16.26 Block diagrams of hybrid beamforming structures at BS for a downlink transmission, where structures A and B denote the full-complexity and reduced-complexity structures, respectively.

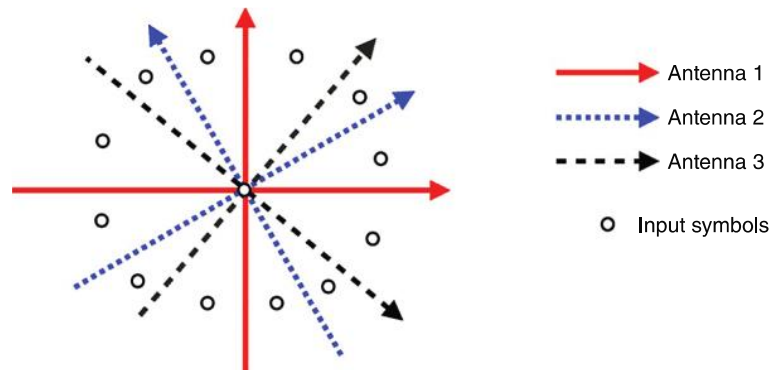


Figure 16.27 Transmitted symbol constellation, and decision regions for three antenna elements. The optimal constellation contains 12 nonzero symbols.

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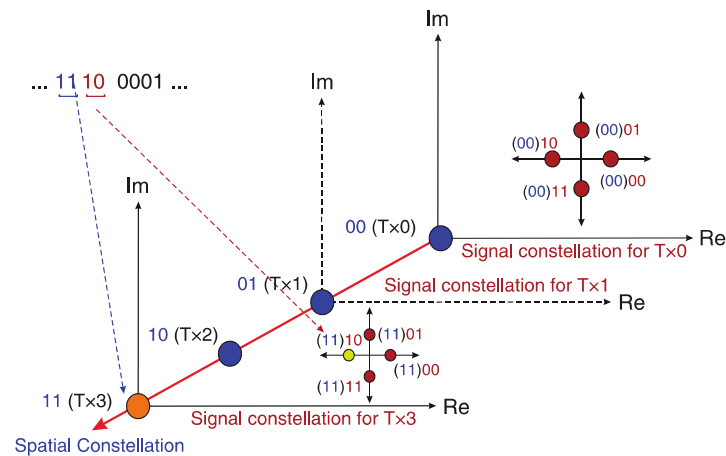


Figure 16.29 Principle of spatial modulation

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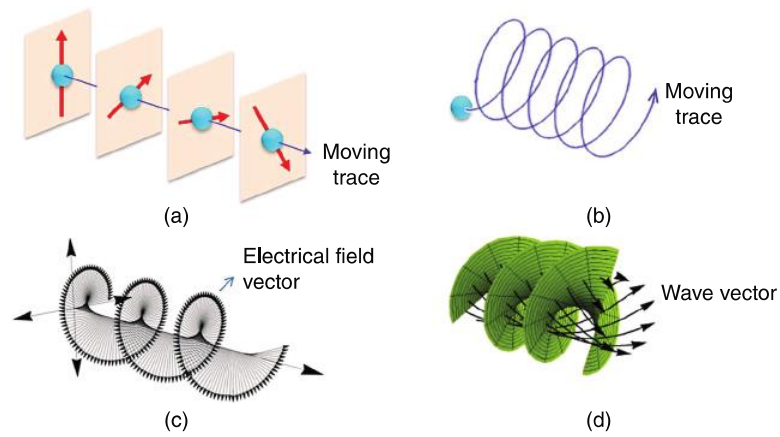


Figure 16.32 Two cases of angular momenta: (a) a spinning object carrying spin and (b) an orbiting object carrying OAM. (c) A circularly polarized wave carrying spin, and (d) the phase structure of an OAM-carrying wave.

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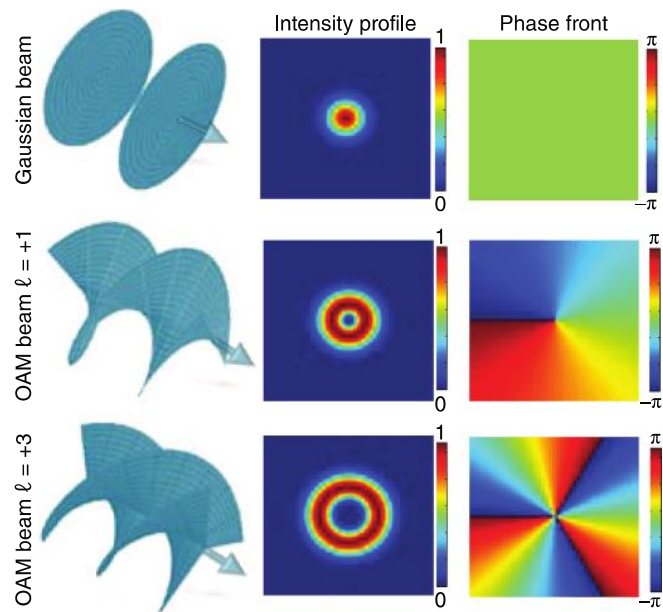


Figure 16.33 Intensity (left) and phase (right) figure for different OAM modes.

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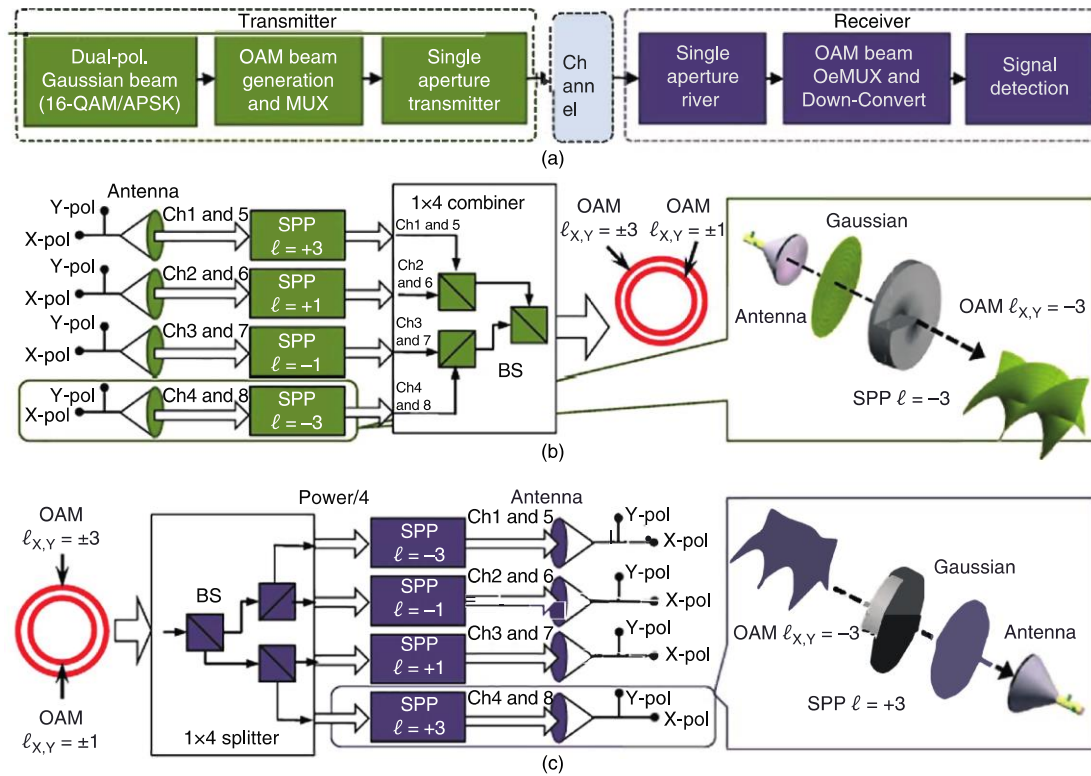


Figure 16.35 System for OAM multiplexing combined with polarization multiplexing. (a) Block diagram, (b) TX multiplexing. (c) RX demultiplexing.

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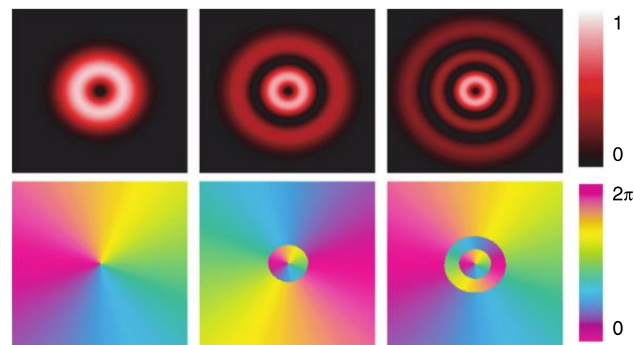


Figure 16.36 Normalized intensity (top) and phase (bottom) plots of Laguerre-Gaussian modes: LG01, LG11, and LG21 (left to right) showing the $p+1$ concentric rings and the effect on the phase pattern.

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Chapter 17

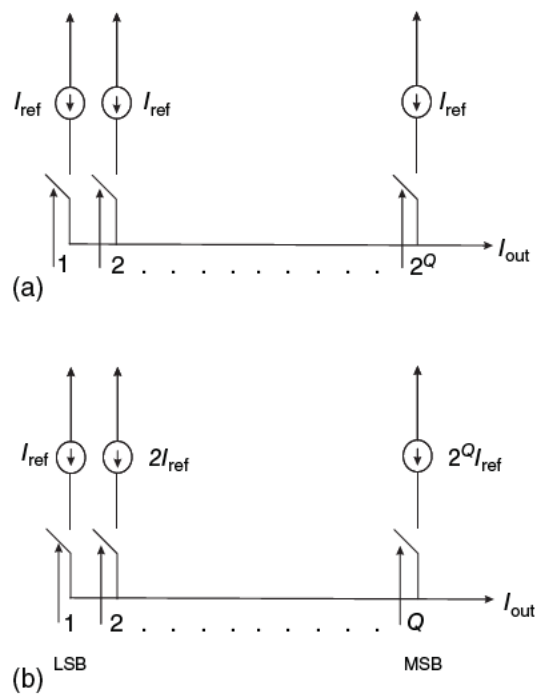


Figure 17.6 Principle of a current array for a DAC: (a) uniform weighting, (b) binary weighting.

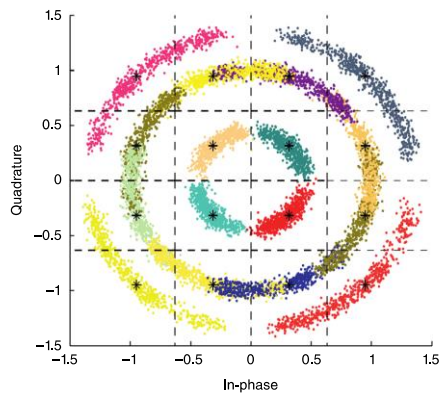


Figure 17.16 16-QAM constellation in the presence of phase noise.
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Chapter 18

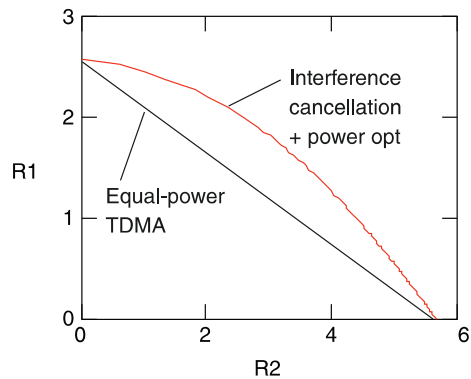


Figure 18.1 Rate/capacity region for two-user broadcast channel: $N_0 = 0.1$, $|h_1|^2 = 0.5$, $|h_2|^2 = 5$, $P = 1$.

Chapter 20

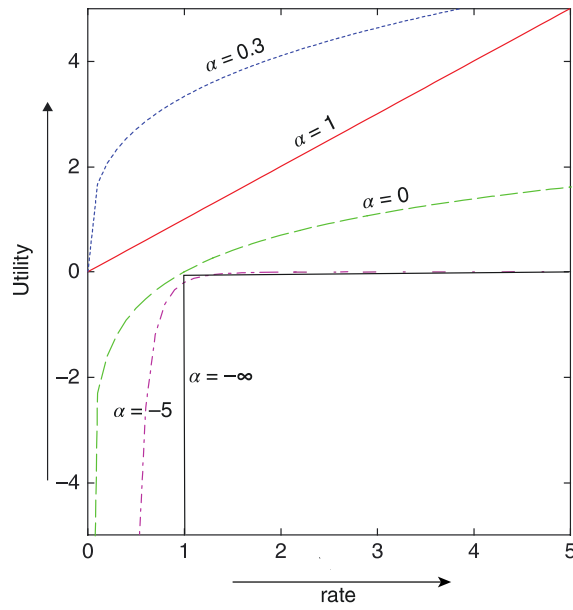


Figure 20.2 α -Fair utility functions with $\alpha = 1, 0.3, 0, -5, -\infty$

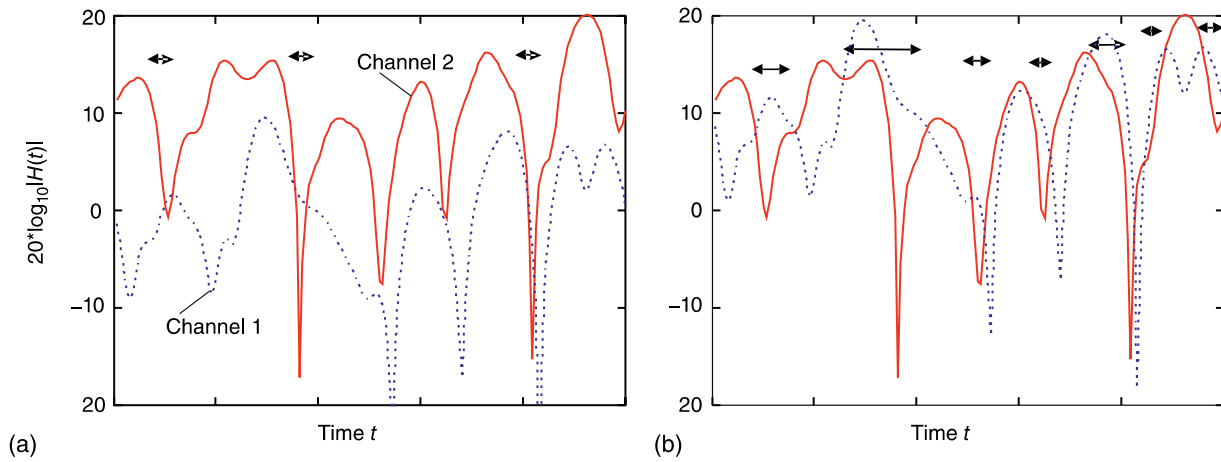


Figure 20.3 SNR as function of time for two users, and the selection of users to get resources for max-rate scheduling (a) and proportional fair (b). Channels are the same, but in figure (b) are normalized to same average power to show principle of proportional fair scheduling. Arrows indicate times at which user with Channel 1 is chosen.

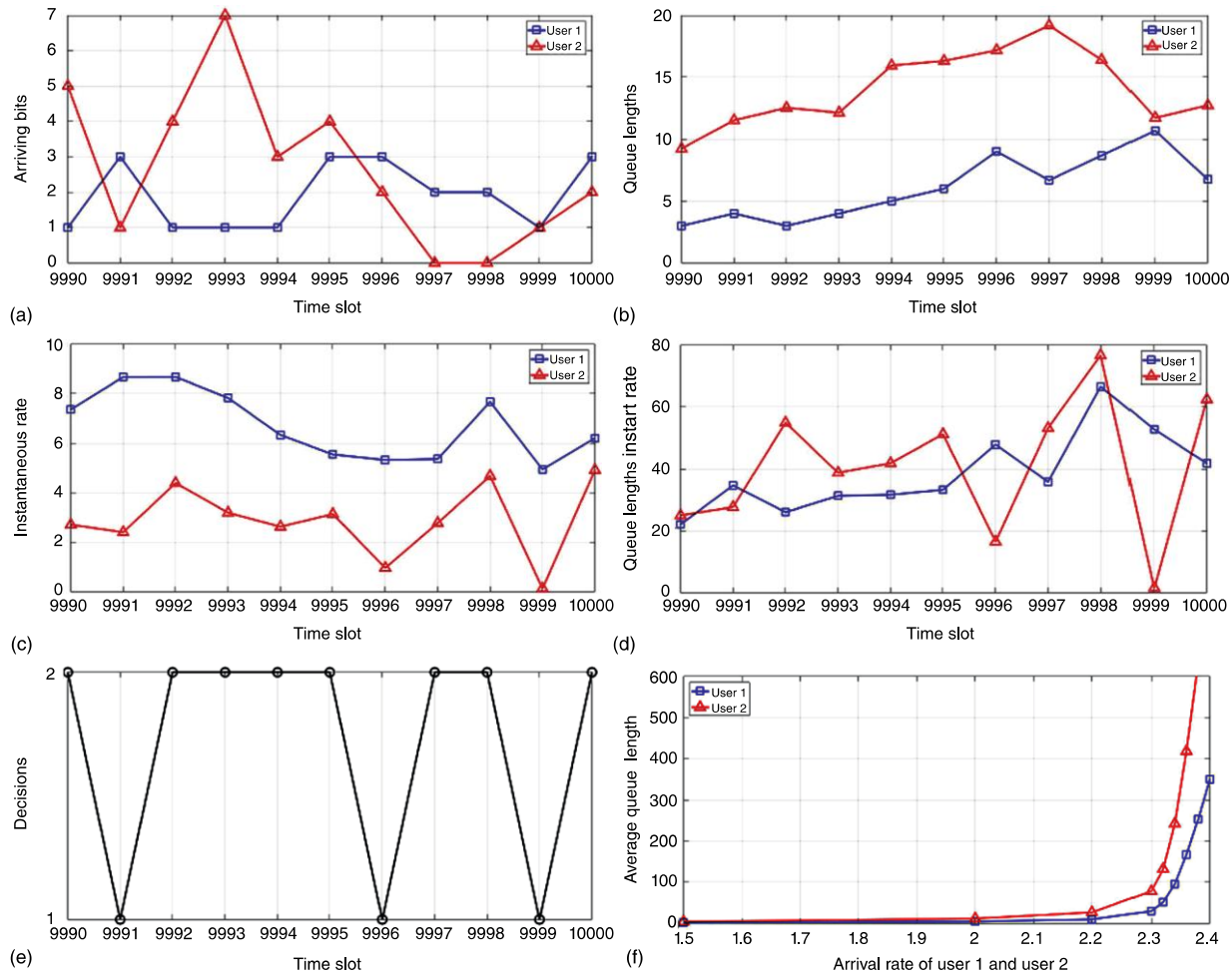


Figure 20.5 Scheduling of a two-user system with random arrivals with arrival rate $\nu = 2$, $\gamma_1 = 20$ dB, $\gamma_2 = 10$ dB. Achievable rate is Shannon capacity; only one user scheduled per timeslot. 20,000 timeslots simulated, results shown for part of that range in steady-state. (a) Number of arrivals. (b) Queue evolution. (c) Instantaneous rate evolution. (d) Evolution of the backpressure (e) Decisions of which user signals to send. (f) Queue length as function of arrival rates of the users. From M.C. Lee, private communication

Chapter 21

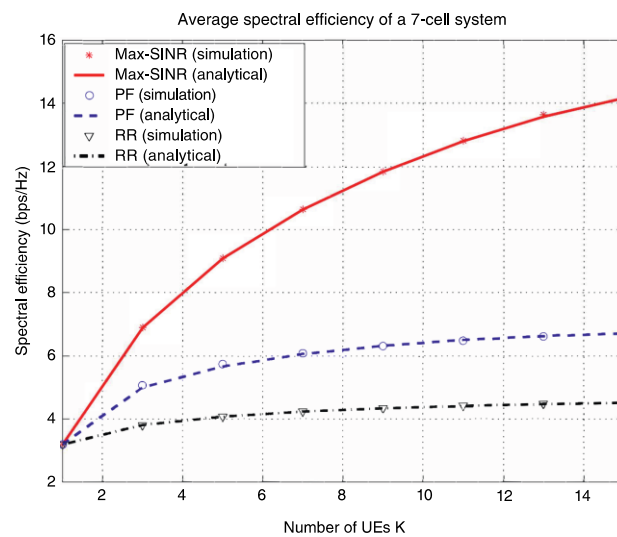


Figure 21.4 Short-term Rayleigh fading-averaged spectral efficiencies of systems with various schedulers. Pathloss exponent is 3.7, the dB standard deviation of all lognormal RVs is $\sigma = 8$ dB. The SNR at the cell corner is $\gamma_{\text{edge}} = 10$ dB. The k th UE is placed at a distance of kR from its serving BS, where R is the cell radius, and at an azimuth of $2k\pi/K$.

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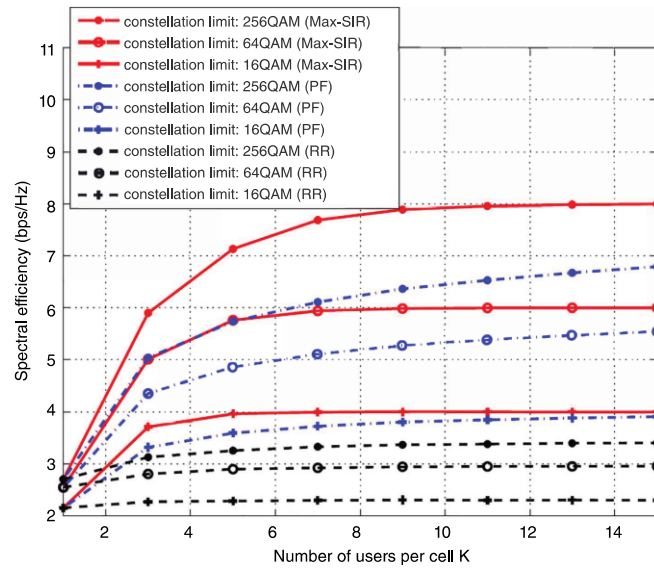


Figure 21.5 Composite channels: effects of constellation size limits on spectral efficiency for RR, PF, and Max-SIR schedulers.

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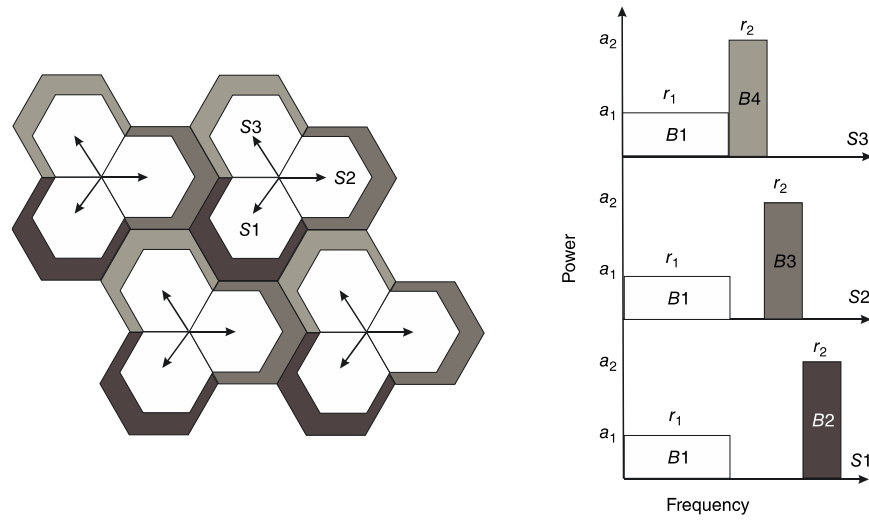


Figure 21.6 Principle of partial frequency reuse and sectorization.

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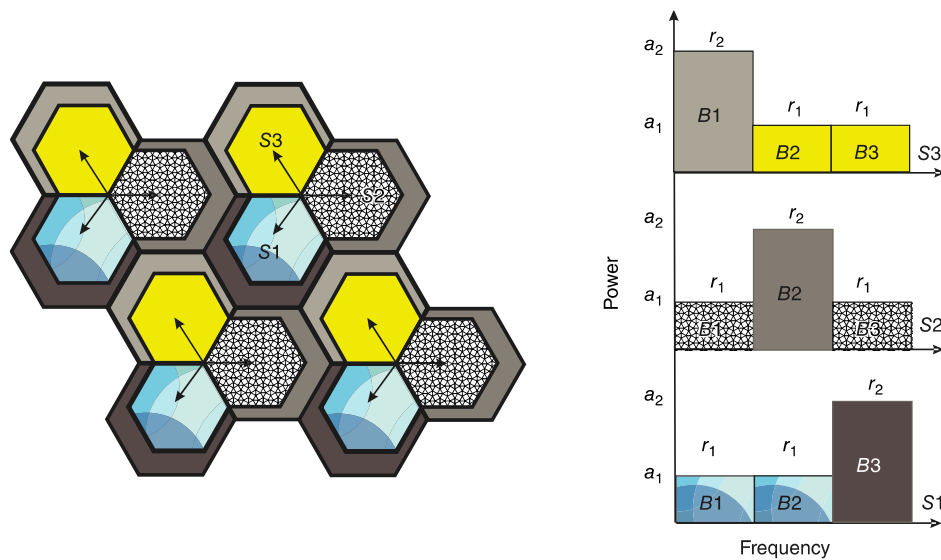


Figure 21.7 Principle of soft frequency reuse.

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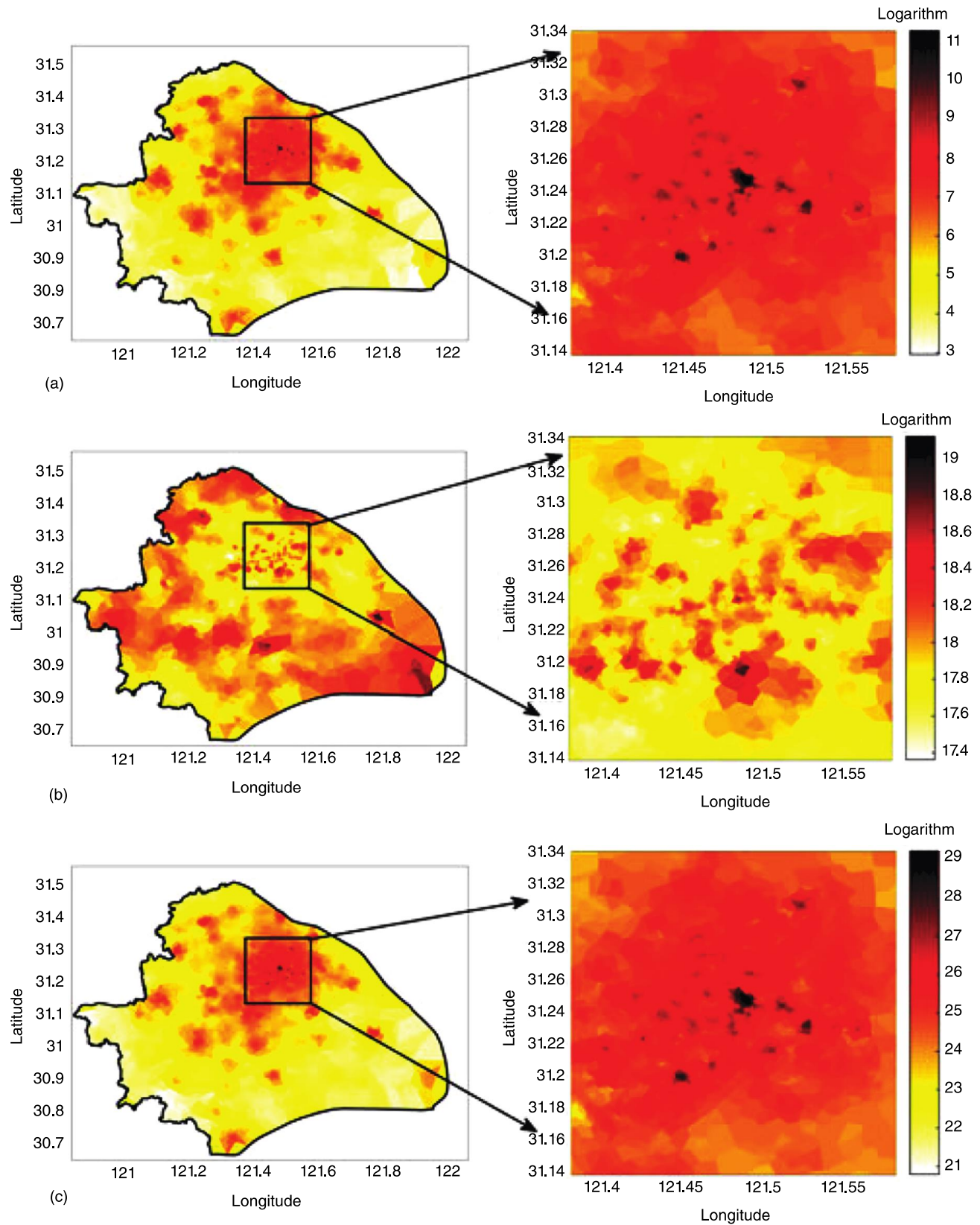


Figure 21.10 Geographical distributions of subscriber density (a), per-subscriber demand (b), and traffic density (c) in Shanghai, China.

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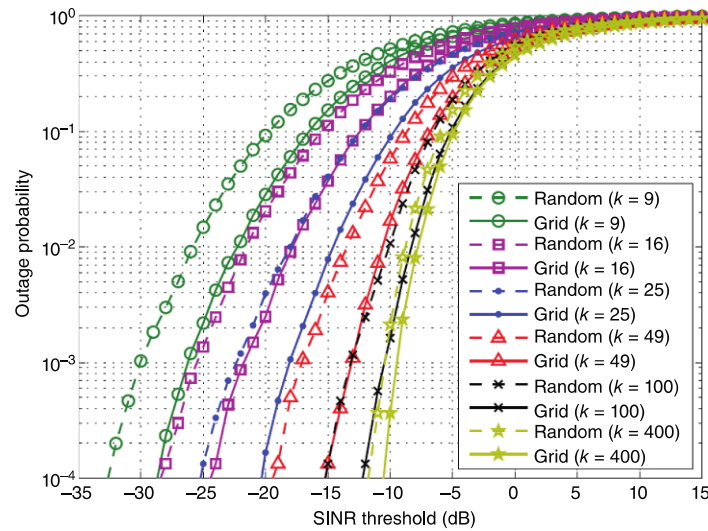


Figure 21.12 Outage probability versus SINR in grid and random topologies. k is the number of BSs per 1 km^2 .

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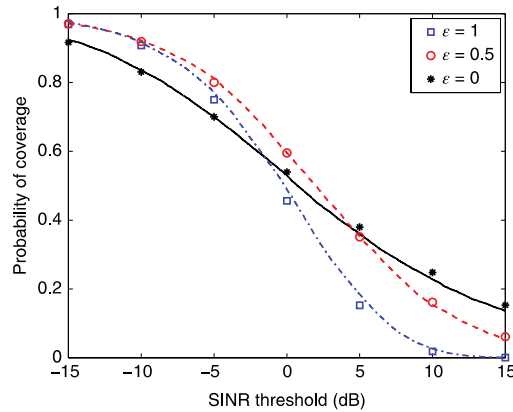


Figure 21.13 Coverage probability as function of required SINR for different power control policy when using stochastic geometry analysis.

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Chapter 22

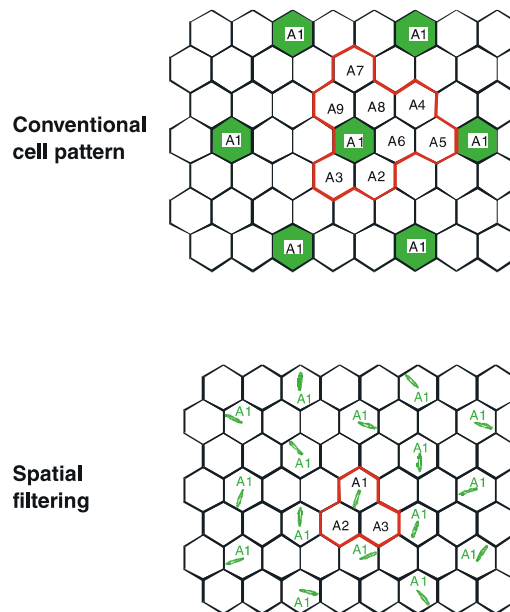


Figure 22.1 Principle of spatial filtering for interference reduction.

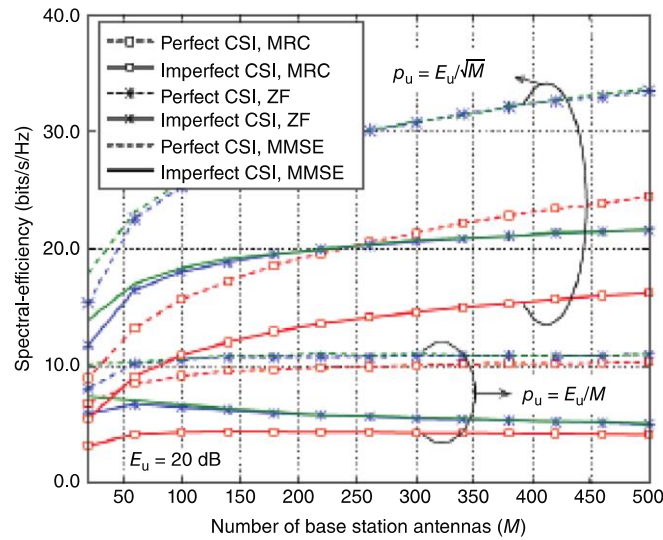


Figure 22.9 Spectral efficiency of massive MIMO versus number of BS antennas. $K = 10$ UEs are served simultaneously. Both the case that transmit power scales as $p_u = E_u / \sqrt{N_{BS}}$ and as $p_u = E_u / N_{BS}$ is shown. Note that M in the figure means N_{BS} in the notation of this book.

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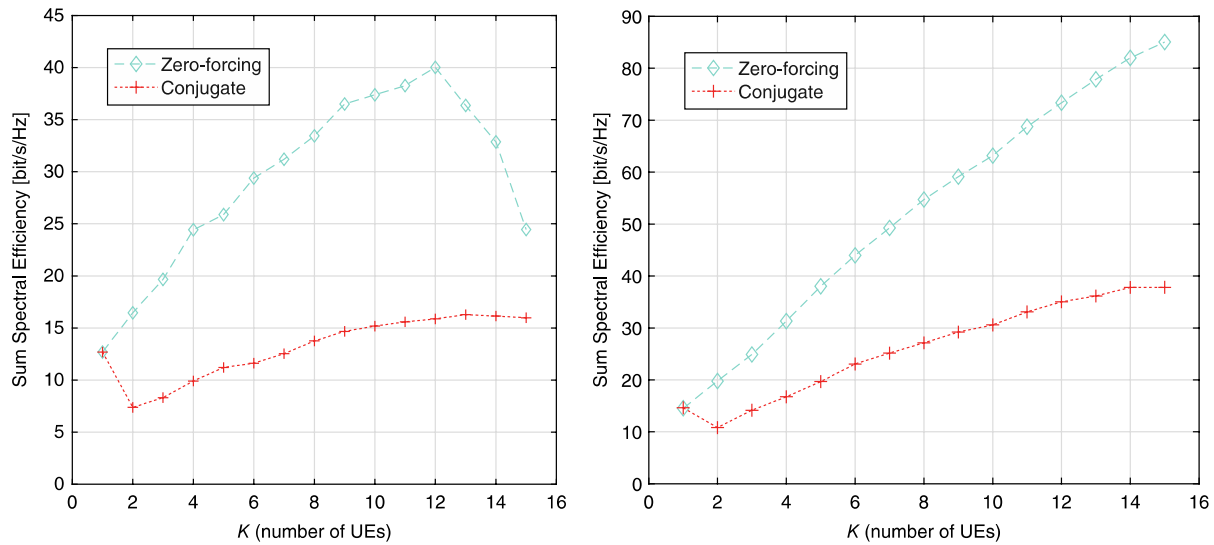


Figure 22.10 Capacity measured in an indoor channel with a massive MIMO testbed (Argos). (a) $N_{BS} = 15$. (b) $N_{BS} = 64$. From C. Shepard, private communication (2021).

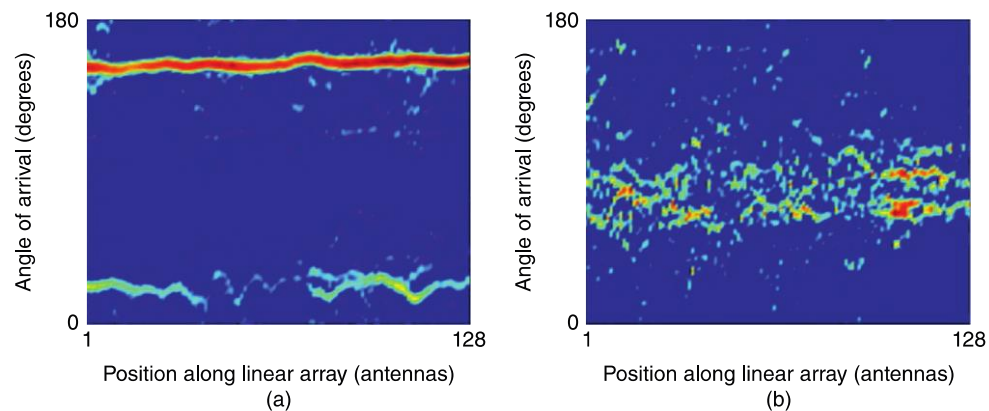


Figure 22.13 Angular power spectrum as a function of the position along a 128-element linear array. (a) A LOS scenario (b) a NLOS scenario.

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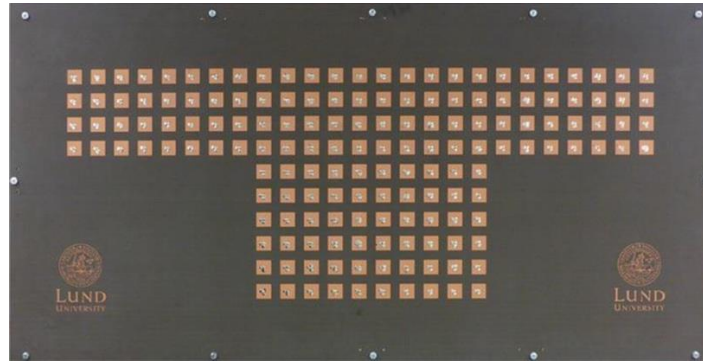


Figure 22.14 Example of two-dimensional array at microwave frequencies (3.7 GHz).

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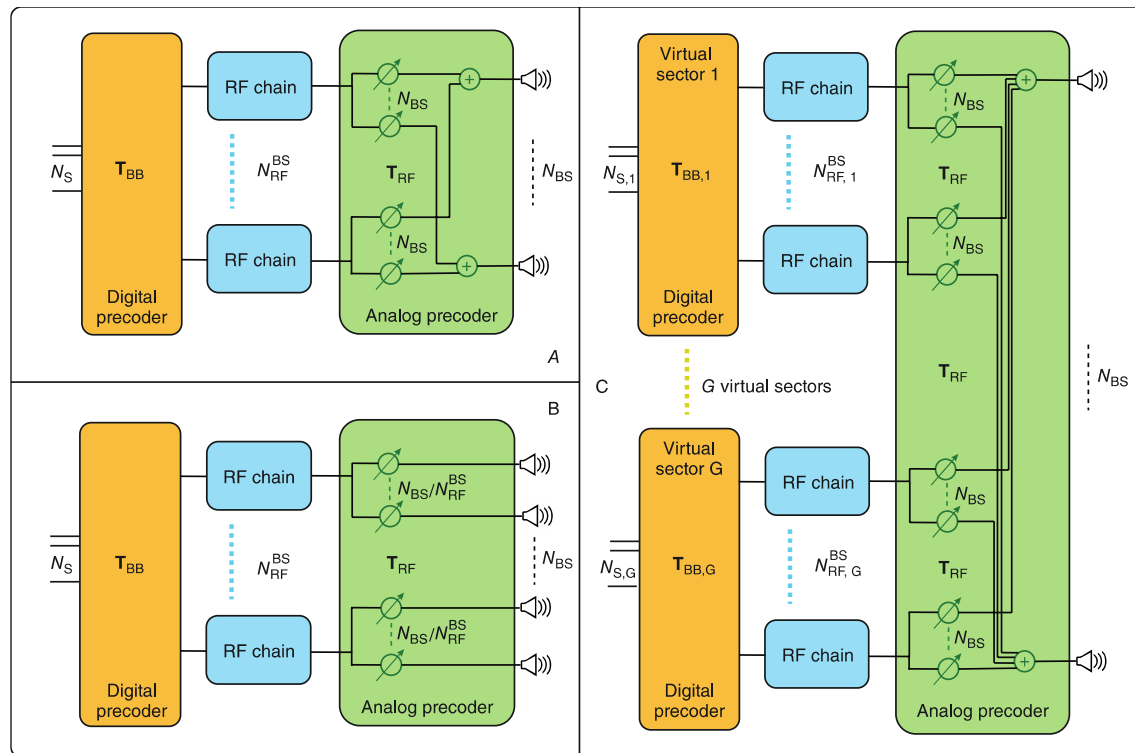


Figure 22.16 Block diagrams of hybrid beamforming structures at BS for a downlink transmission, where structures A, B, and C denote the full complexity, reduced-complexity, and virtual-sectorization structures, respectively.

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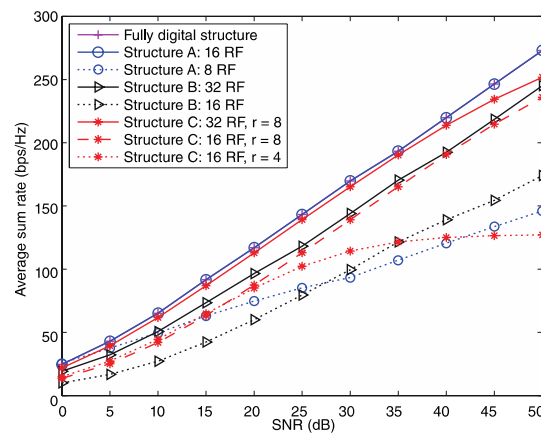


Figure 22.17 Performance comparison of the three hybrid structures with MU-MIMO; $N_{BS} = 64$, $N_{UE} = 1$, four groups of users located in a sector with mean directions $[-45^\circ, -15^\circ, 15^\circ, 45^\circ]$, and each group has four users.

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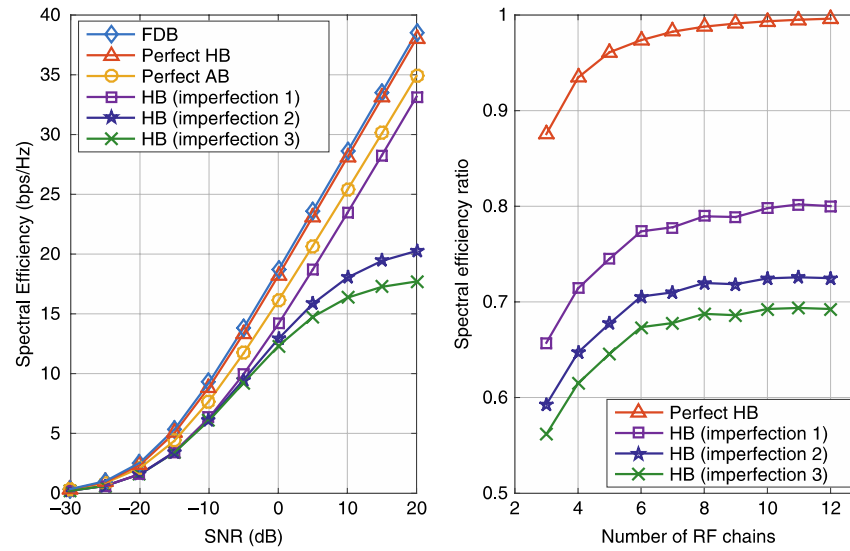


Figure 22.18 SE comparison of Fully-Digital Beamforming (FDB), Hybrid Beamforming with perfect RF hardware (Perfect HB), Analog-only beam steering with perfect RF hardware (Perfect AB), and HB with three different cases of RF hardware imperfection: (a) quantization error caused by limited-resolution phase shifters; (b and c) additional residual transceiver impairments at BS and at both BS and UE, respectively. Right subfigure, spectral efficiency normalized to FDB.

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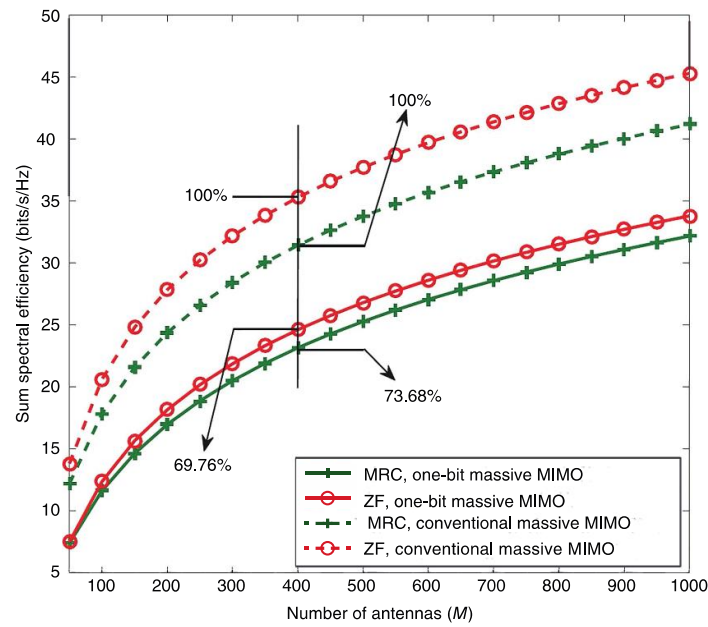


Figure 22.20 Comparison of uplink sum spectral efficiency versus number of BS antennas, with average SNR $\gamma = -10$ dB.

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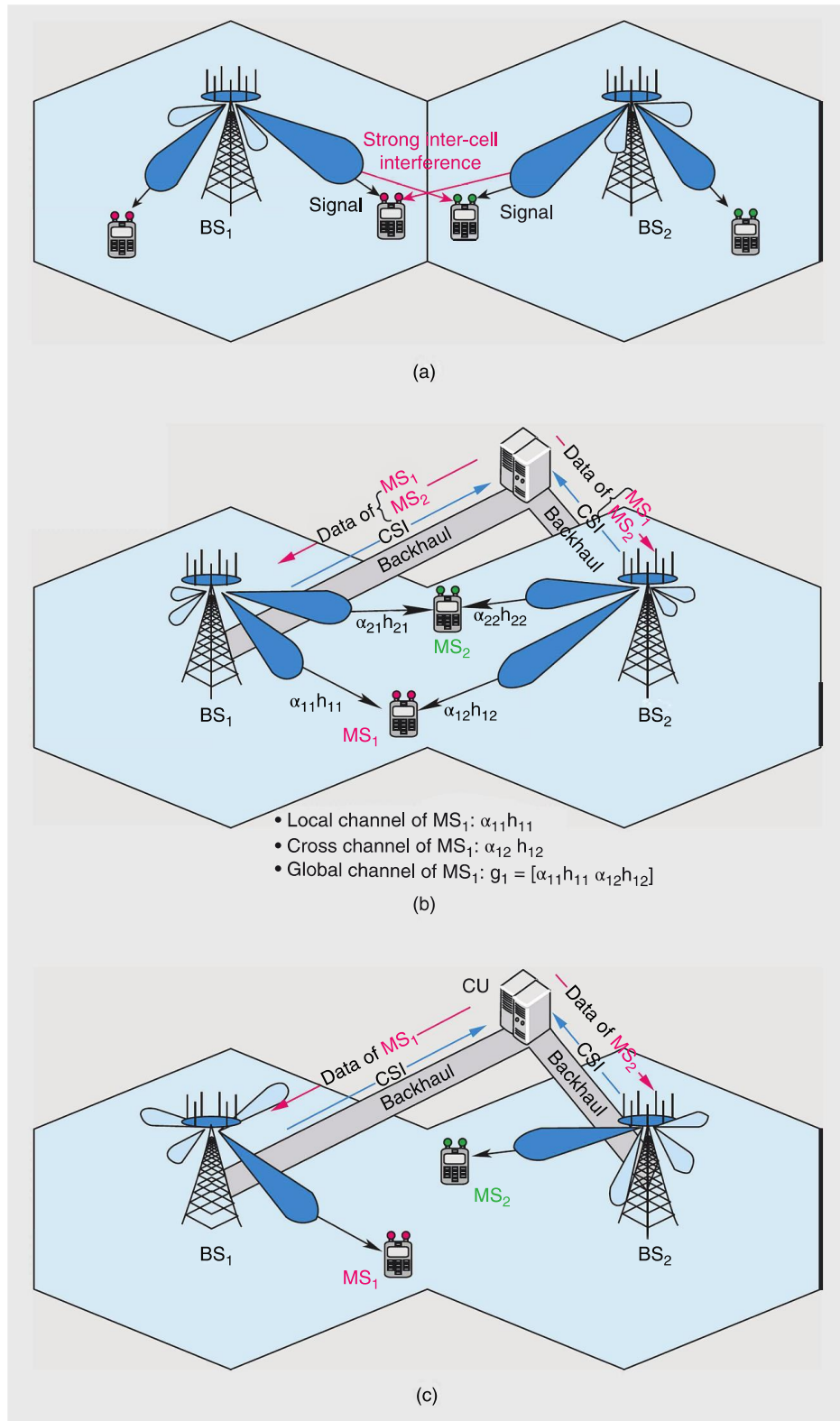


Figure 22.21 Illustration of Non-CoMP, CoMP-JP, and CoMP-CB. For the CoMP-CB systems, BS_1 is the serving BS of MS_1 , from which the average channel gain is stronger. (a) Non-CoMP: no coordination between BSs, and strong interference might be caused to adjacent cell edge users. (b) CoMP-JP: BSs cooperate by jointly serving multiple users in their covered area, (c) CoMP-CB: BSs cooperate by avoiding interference to adjacent celledge users. The acronym MS in the figure is UE in the notation of this book.

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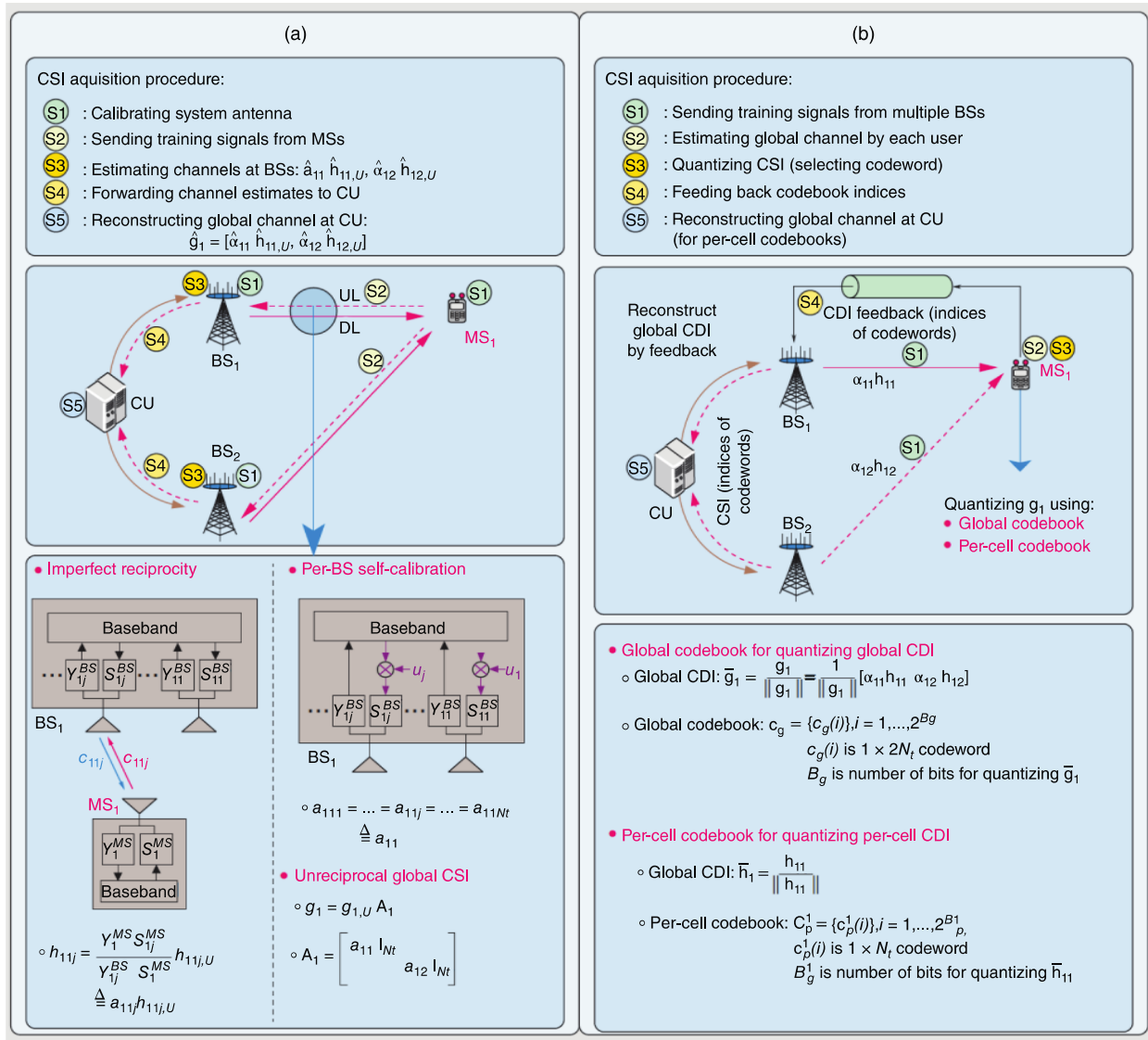


Figure 22.22 CSI acquisition procedures in TDD (a) and FDD (b) CoMP systems, where the specific steps from S1 to S5 are respectively shown in each of the procedure. The acronym MS in the figure is UE in the notation of this book.

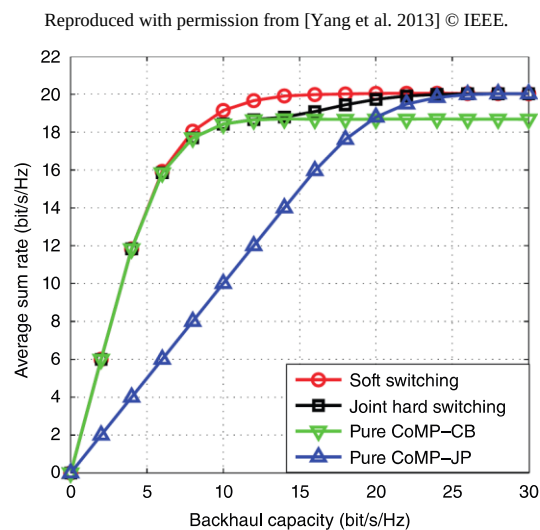


Figure 22.23 Average sum rate achievable in a ComP system with finite backhaul capacity, comparing pure CoMP-CB, pure CoMP-JP, and switching between the two.

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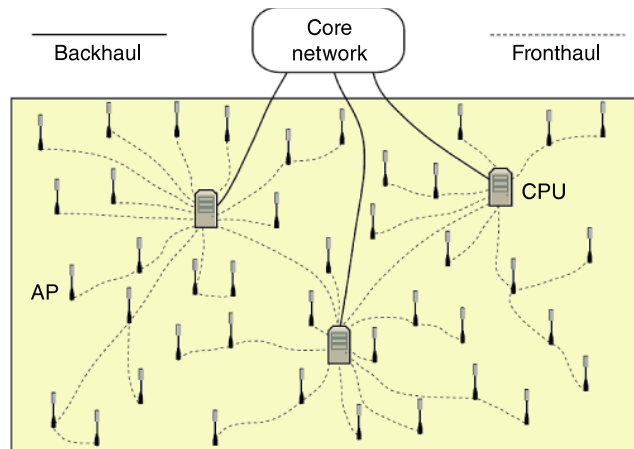


Figure 22.24 Architecture of a CF-mMIMO system.

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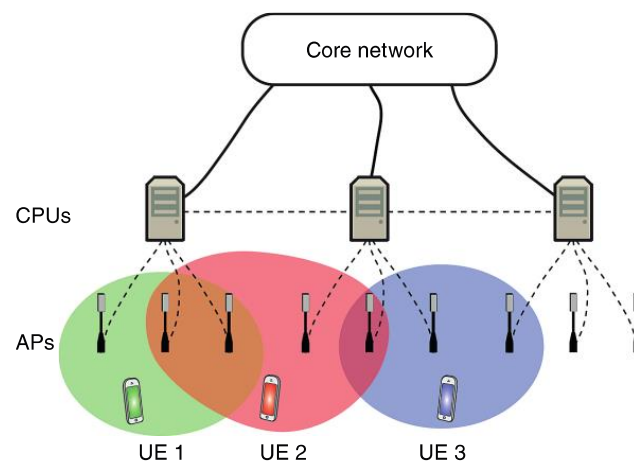


Figure 22.25 Principle of dynamic AP association in a cell-free network. Each UE is associated with the APs in its shaded region.

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Chapter 23

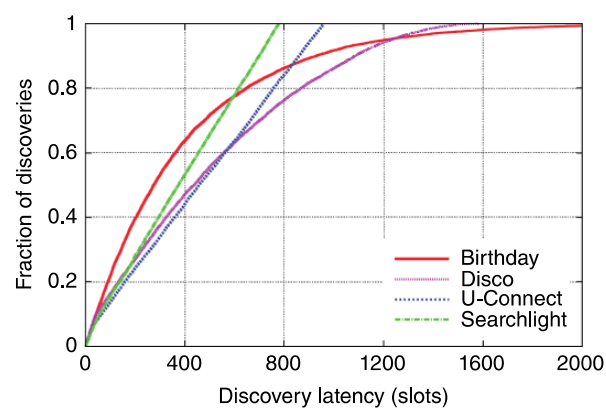


Figure 23.4 Latency of discovery with various methods with 5% duty cycle.

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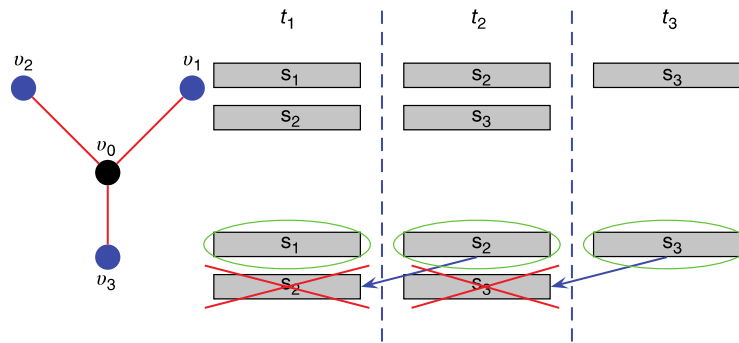


Figure 23.5 Operating principle of Zig-Zag discovery

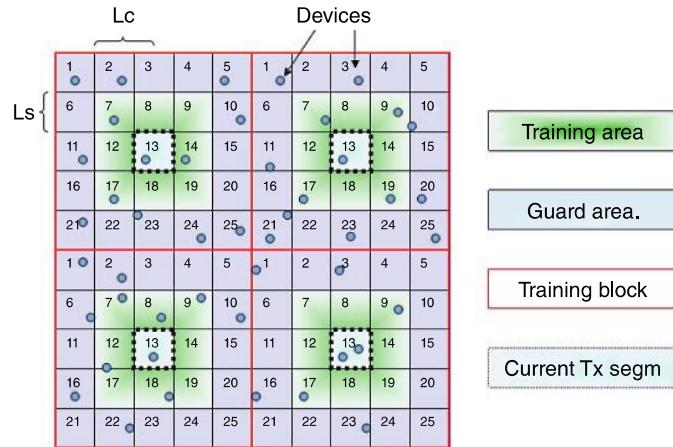


Figure 23.6 Simplified segmented cell and LATS cell components. Devices in training set 13 are allowed to transmit.

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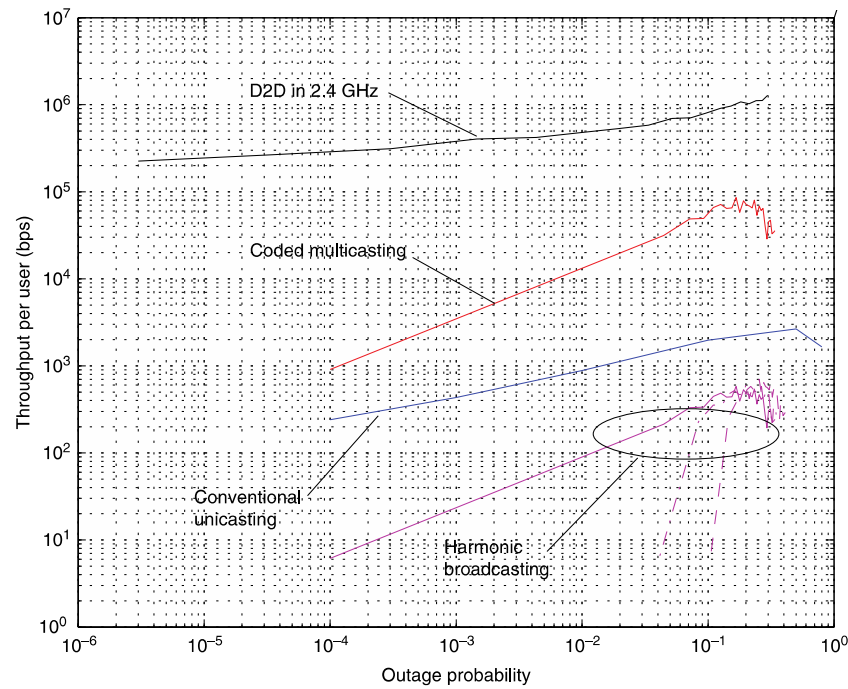


Figure 23.20 Simulation results for the throughput-outage trade-off for conventional unicasting, coded multi-casting, harmonic broadcasting (with different numbers of transmitted files), and the caching-plus-D2D-communication scheme under 2.4 GHz office channel models.

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Chapter 25

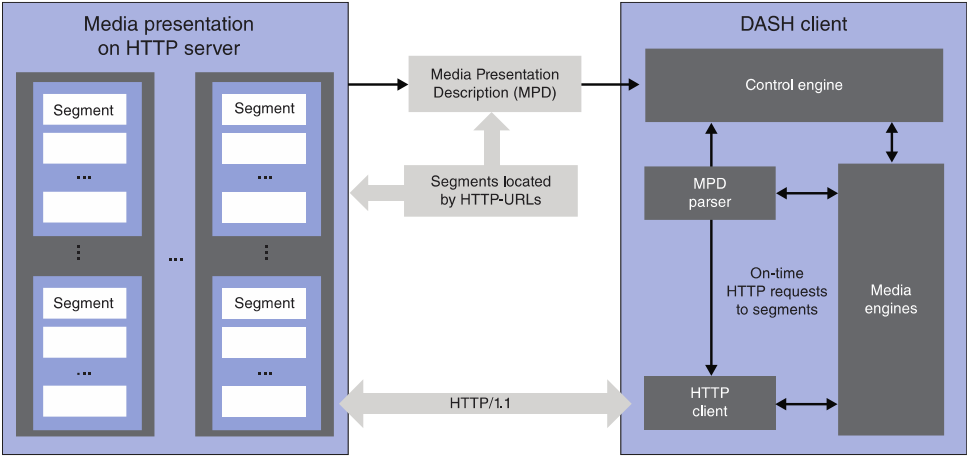


Figure 25.17 Illustration of DASH-based media delivery framework, including media presentation on HTTP composed of segments, media presentation description, and key functions of the DASH client.

Chapter 27

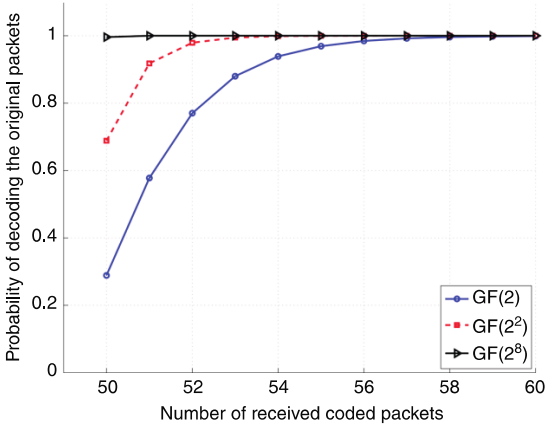


Figure 27.10 For 50 source packets, number of packets that need to be transmitted with random network coding for Galois fields of various sizes. Color version available at wiley.com/go/molisch/wireless3e.

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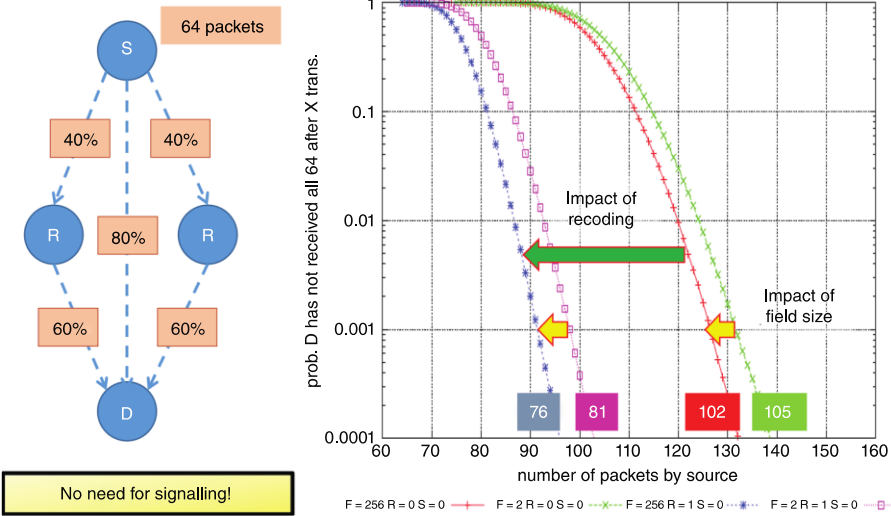


Figure 27.11 Number of required transmissions for sending 64 packets over the shown network with and without recoding.

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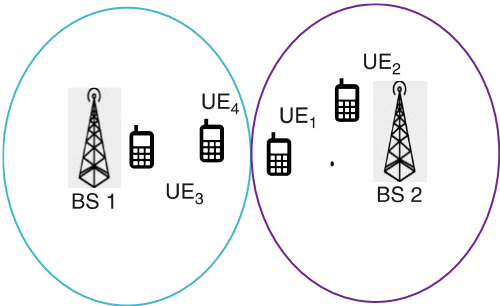


Figure 28.8 Inter-cell interference for NOMA.

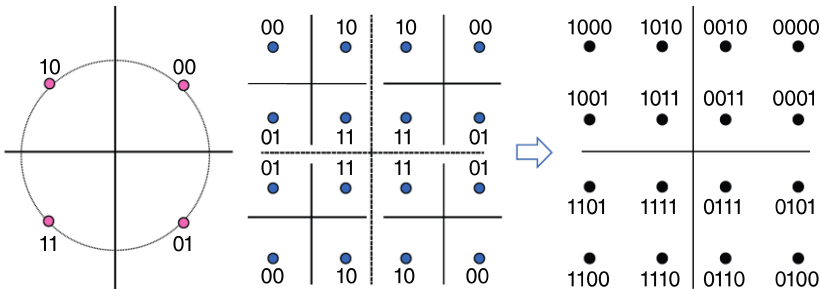


Figure 28.9 Gray encoding for two-user NOMA. Each user uses QPSK; power ratio is 4:1.

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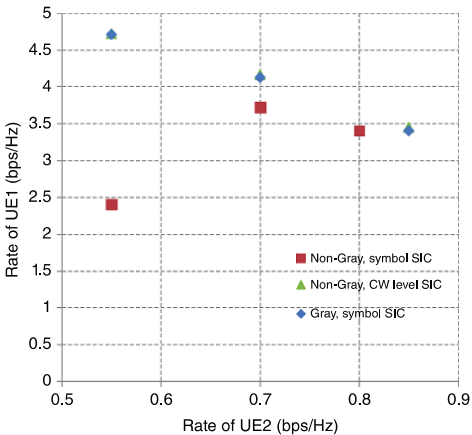


Figure 28.10 Achievable rates for two-user NOMA SNR=20 dB for UE1, SNR =0 dB for UE2.

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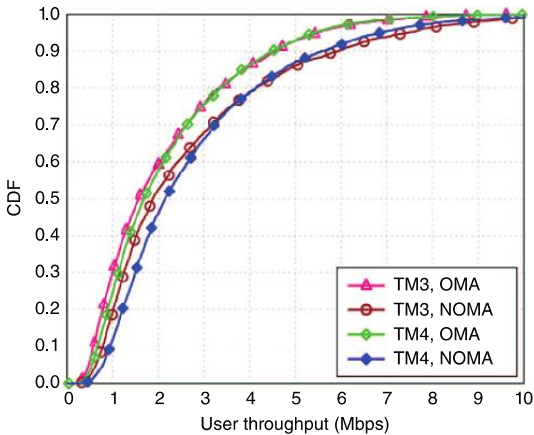


Figure 28.11 Simulations comparing NOMA with OMA with 2 × 2 MIMO; TM3: open loop; TM4: closed loop.

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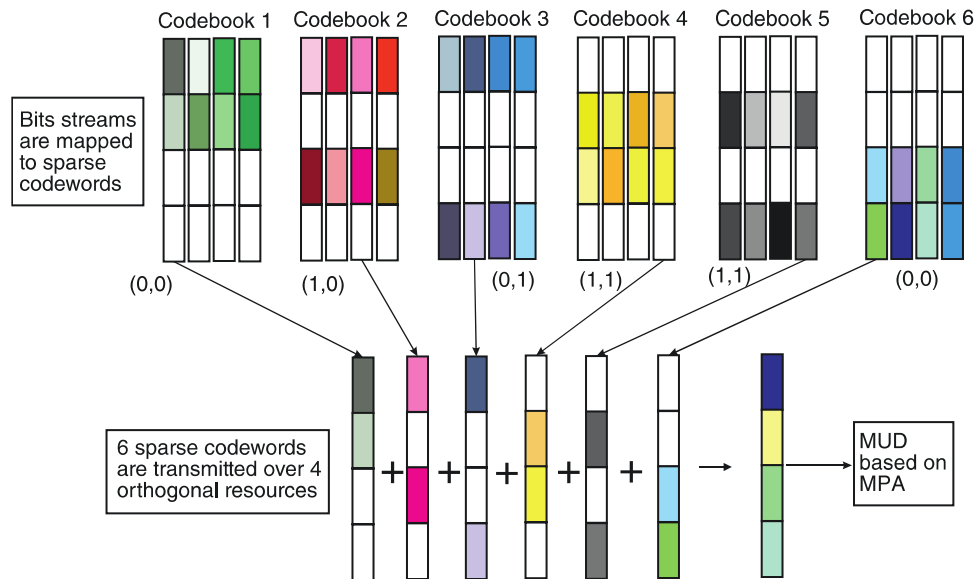


Figure 28.12 Principle of SCMA.

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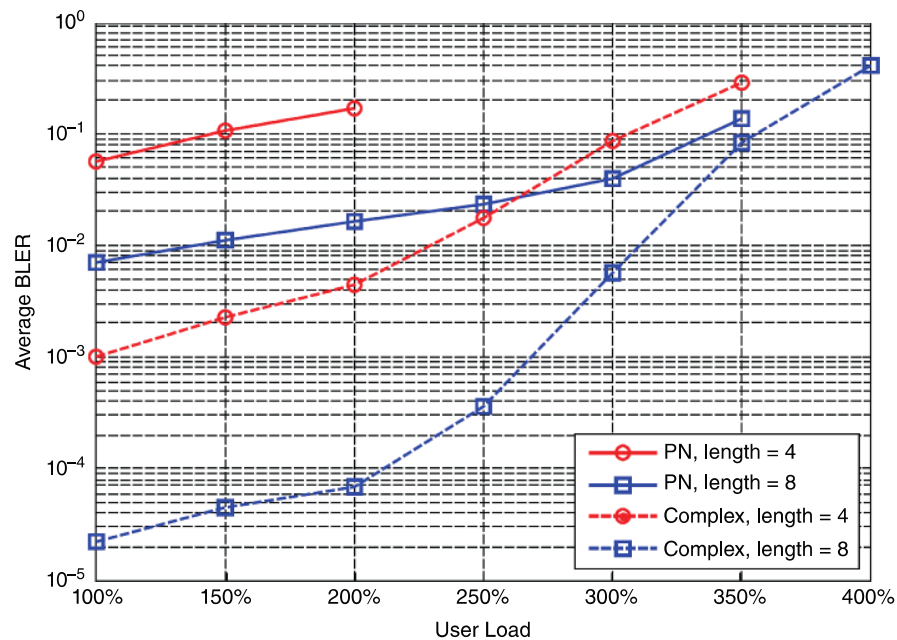


Figure 28.13 Example performance (Block Error Rate, BLER) of MUSA.

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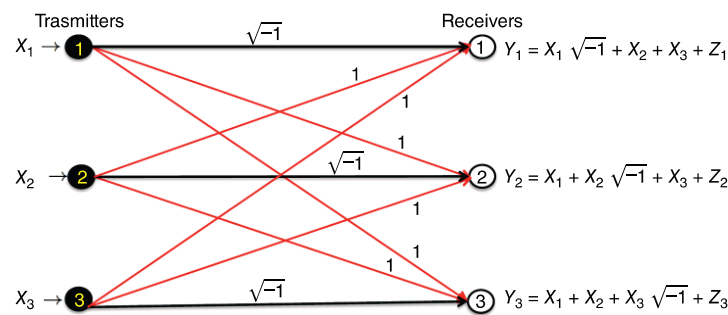


Figure 28.14 Channel in which interference alignment occurs.

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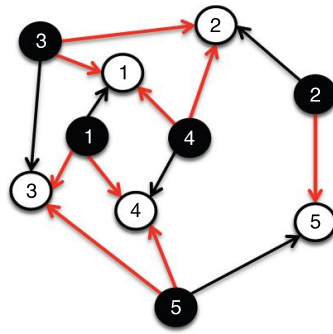


Figure 28.17 Partially connected network. Black circles: TXs. White circles: RXs. Black arrows: desired connections. Red arrows: interference.
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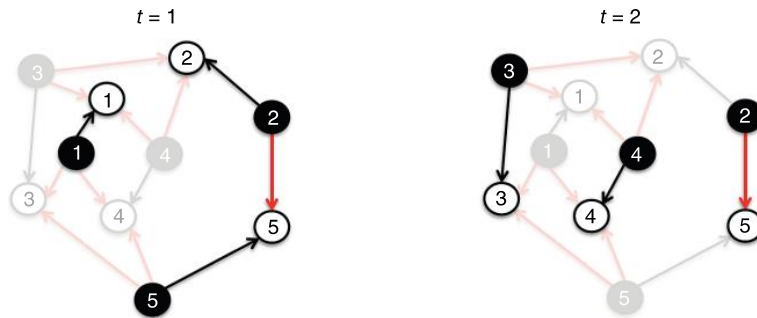


Figure 28.18 Scheduling in partially connected network. Two channel uses (timeslots) are required for five data transmissions.
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Chapter 29

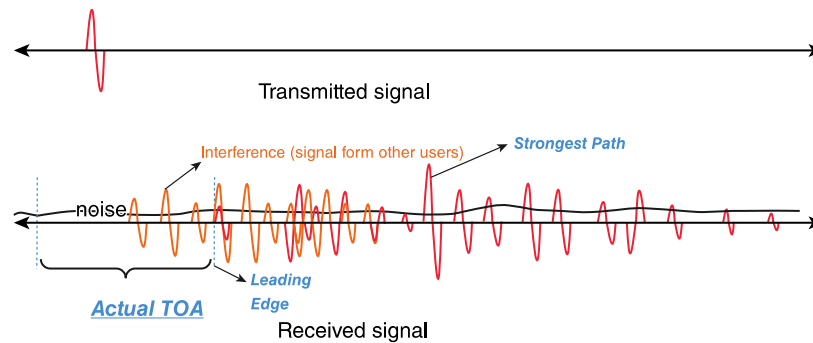


Figure 29.1 Various types of error sources in ranging measurements.

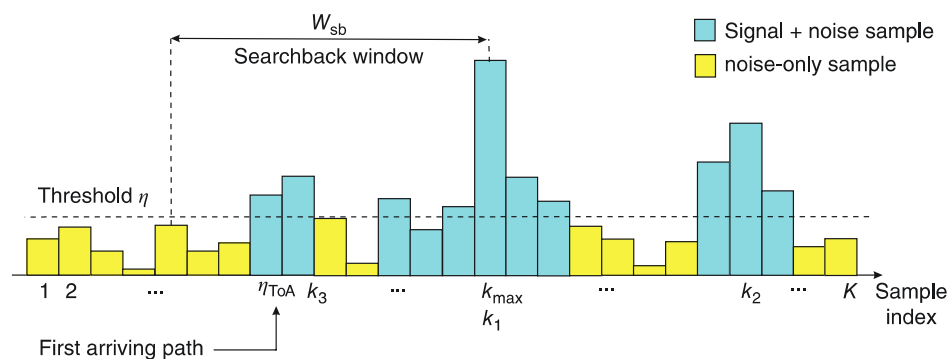


Figure 29.4 Principle of searchback algorithm and thresholding algorithm.

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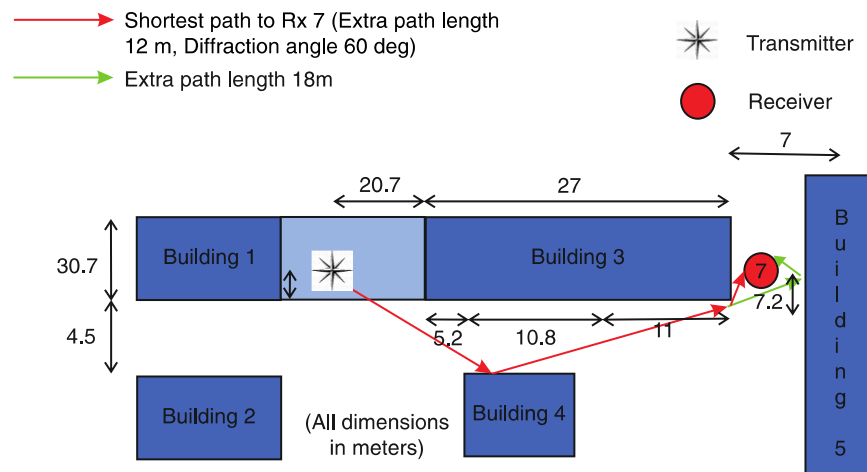


Figure 29.5 Outdoor geometry for measurements.

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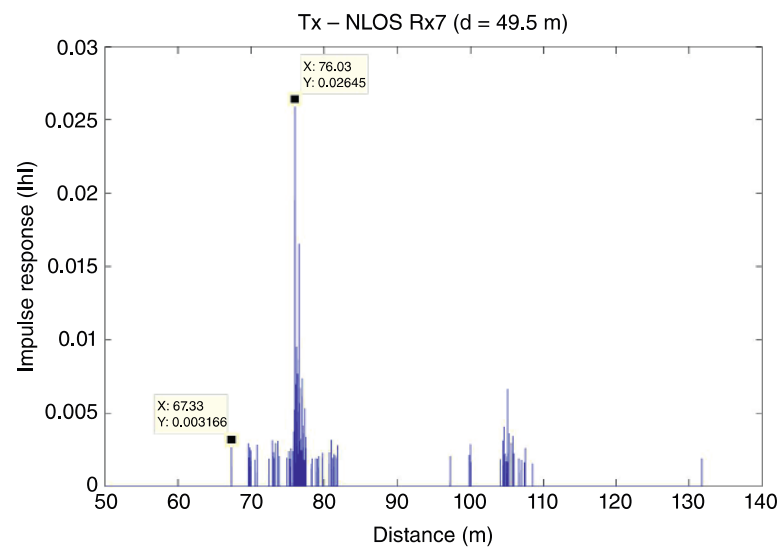


Figure 29.6 Impulse response measured at position 7 of Figure 29.5. Note the large error (nominal distance 49.5 m, pseudorange of first arriving MPC 67 m due to blockage by building; strongest MPC at 76 m).

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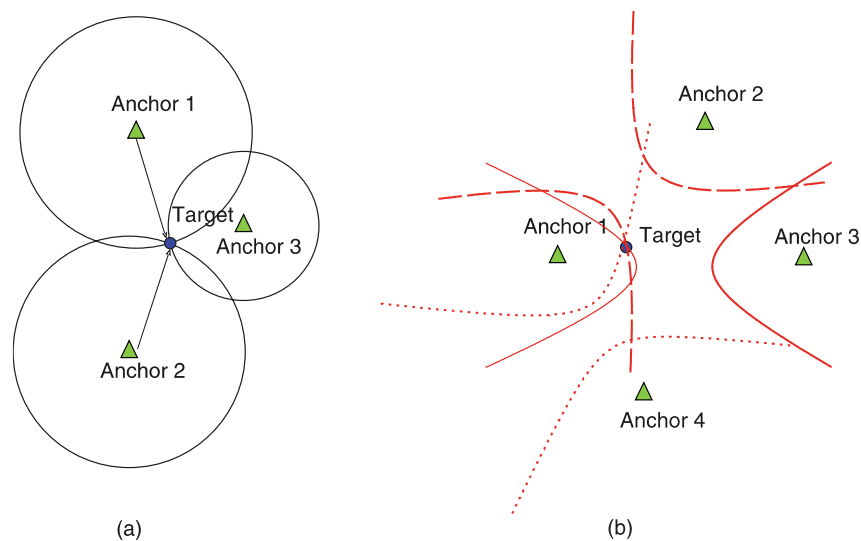


Figure 29.8 Principle of TOA localization (a) and TDOA localization (b).

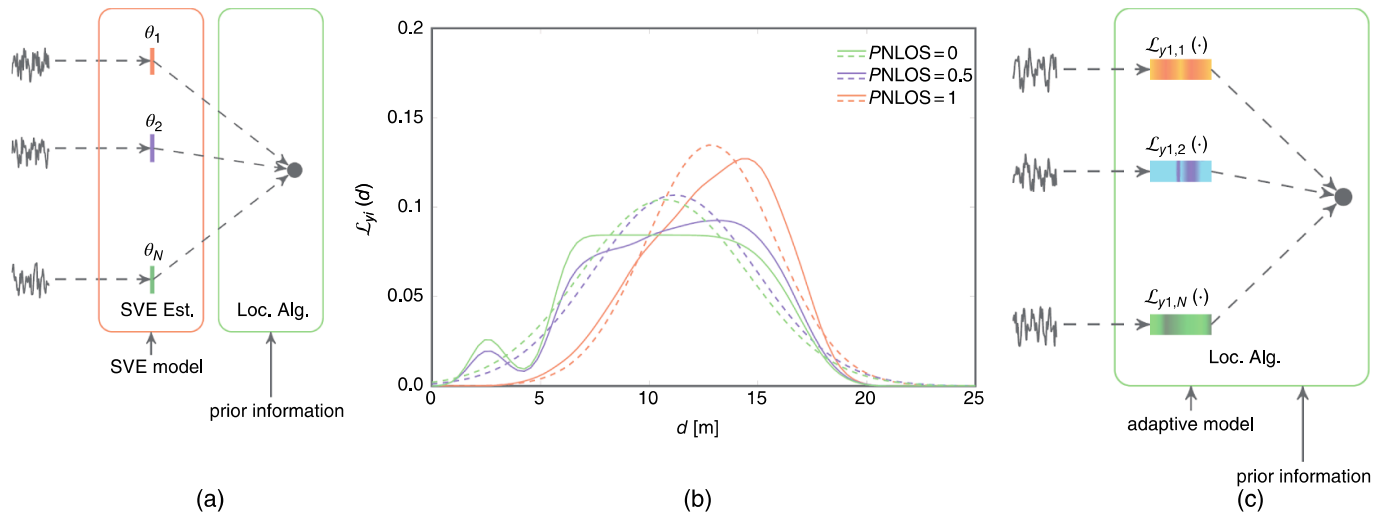


Figure 29.10 Extraction of location by hard thresholding or range estimates (a); likelihood functions of range estimate (b), and location estimate from combining likelihood functions (c).

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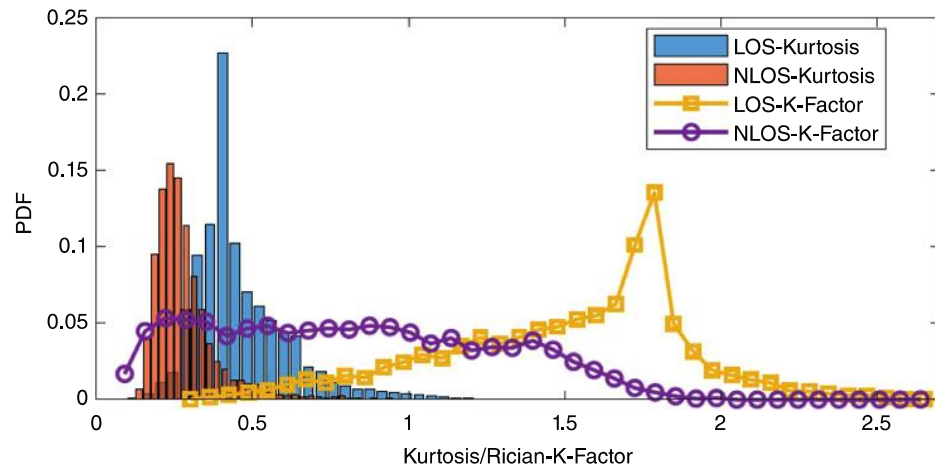


Figure 29.11 Distribution of Rice factor and Kurtosis in LOS and NLOS situations in a vehicle-to-vehicle scenario. It can be seen that in neither case there exists a threshold that allows clear distinction of LOS/NLOS based on one parameter.

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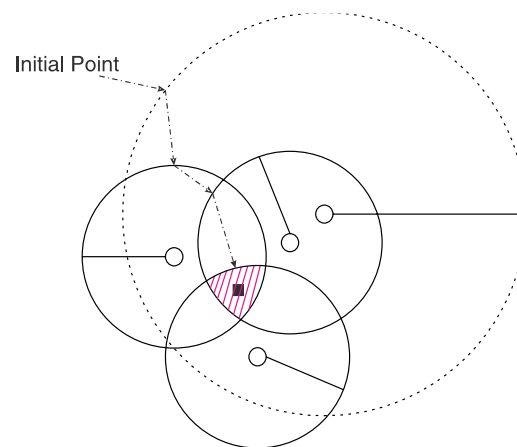


Figure 29.12 POCS is able to suppress a very large positive bias.

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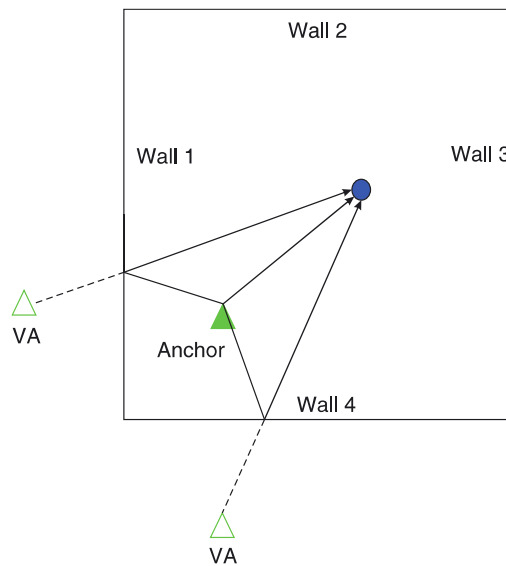


Figure 29.13 Virtual anchors for improved localization.

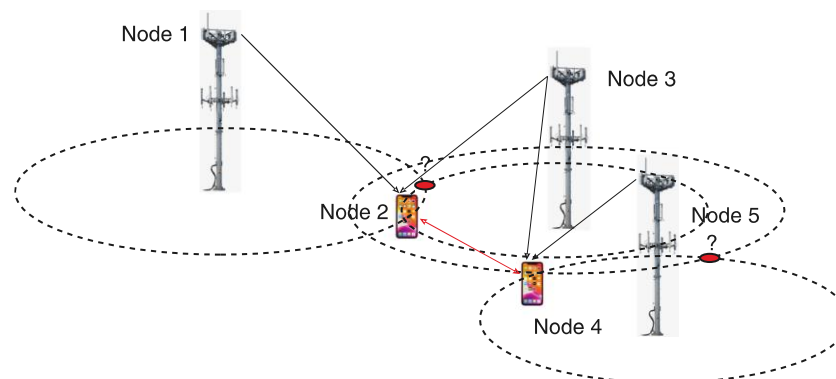


Figure 29.19 Principle of cooperative localization.

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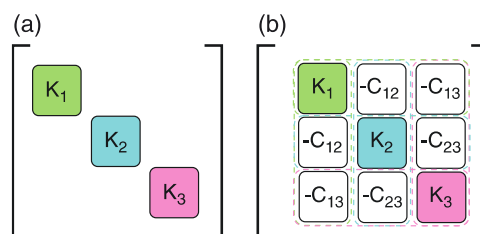


Figure 29.20 Structure of Equivalent Fisher Information Matrix for noncooperative (a) and cooperative (b) case.

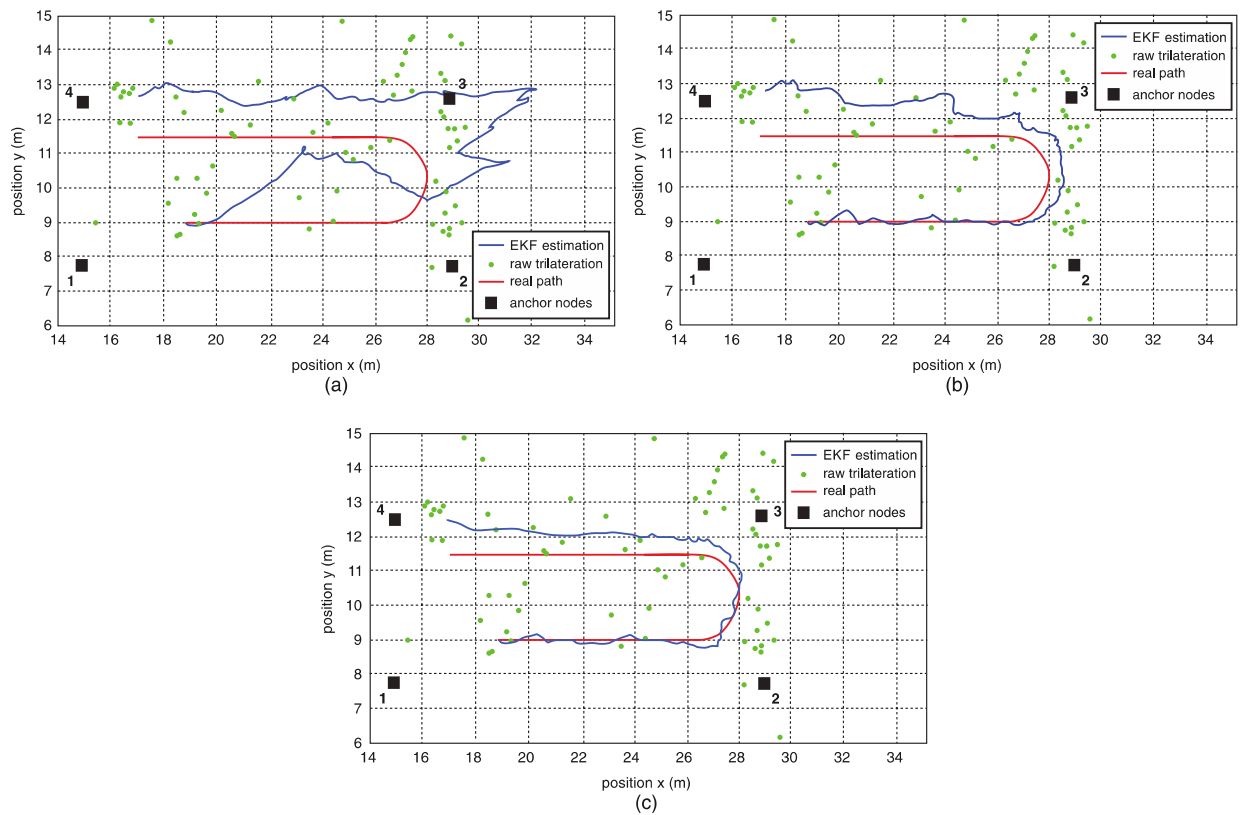


Figure 29.22 EKF tracking in a scenario with some NLOS locations. Black dots signify anchor locations, the black curve is the ground truth trajectory of the agent. Grey points are snapshot-based locations, the grey curve is the EKF output, for (a) standard EKF, (b) omission of all NLOS points, and (c) biased EKF. Reproduced with permission from [Roehrig and Mueller 2009] © IEEE.

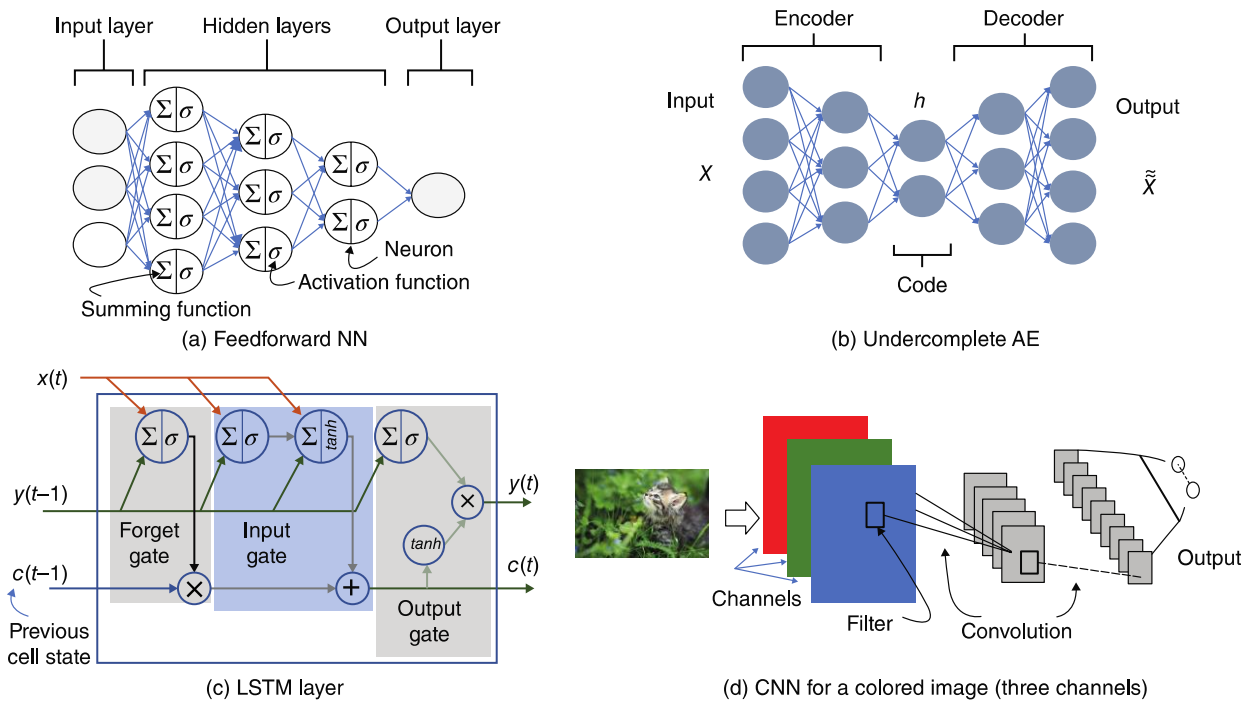


Figure 29.23 Examples of different neural network architectures. Reproduced with permission from [Burghal et al. 2021] Cat image from imazite/Pixabay.

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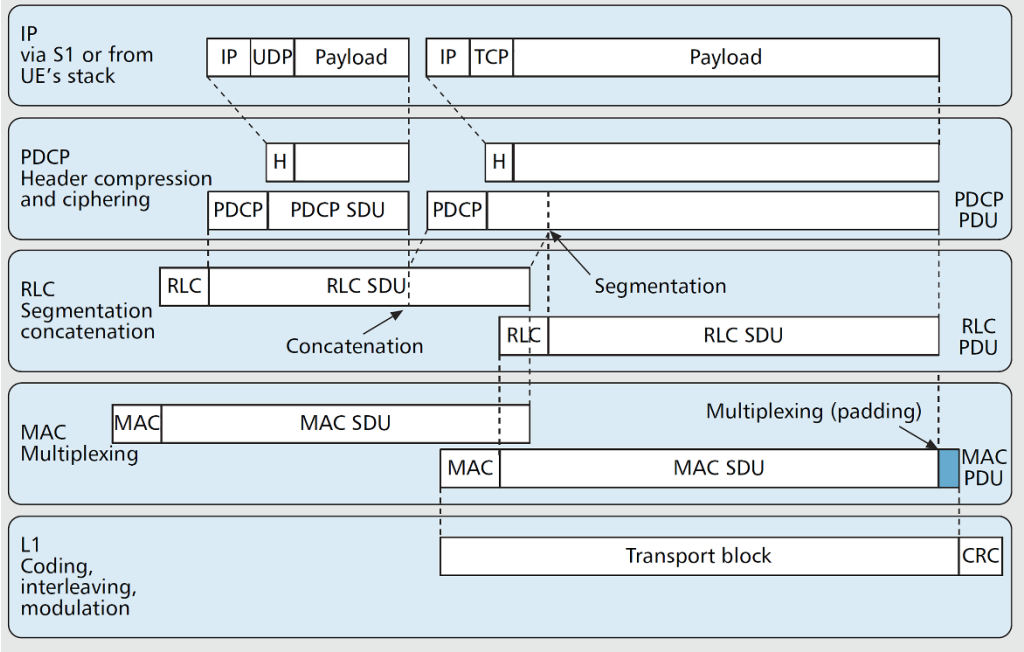


Figure 31.3 Data flow through the protocol stack in LTE.
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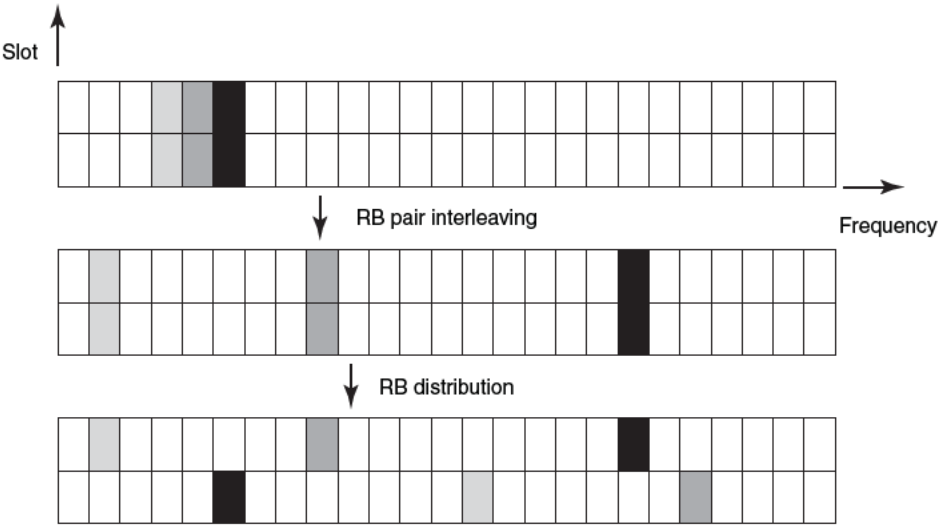


Figure 31.11 Mapping from VRBs to PRBs.
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Figure 32.1 Use cases for 5G envisioned by ETSI. From [ETSI 2020, <https://www.etsi.org/technologies/5g>].

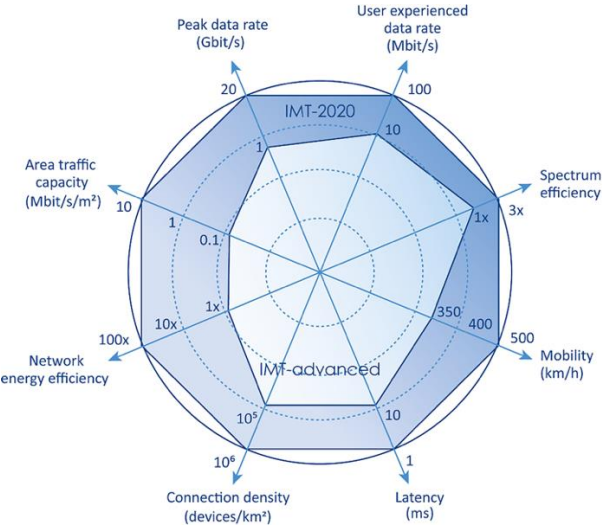


Figure 32.2 Goals of 5G (IMT-2020) compared to 4G (IMT-Advanced).

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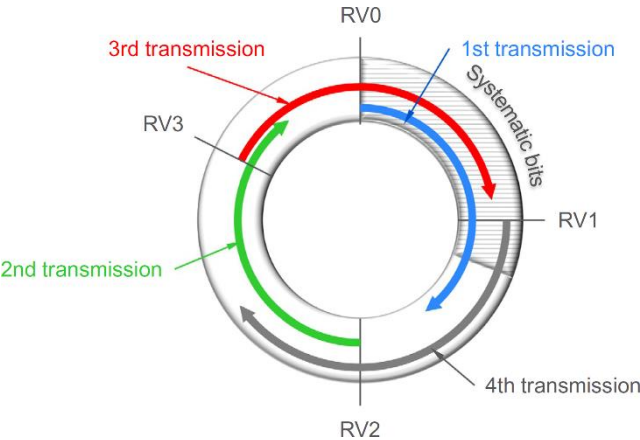


Figure 32.7 Use of the circular buffer to generate different incremental redundancy versions.

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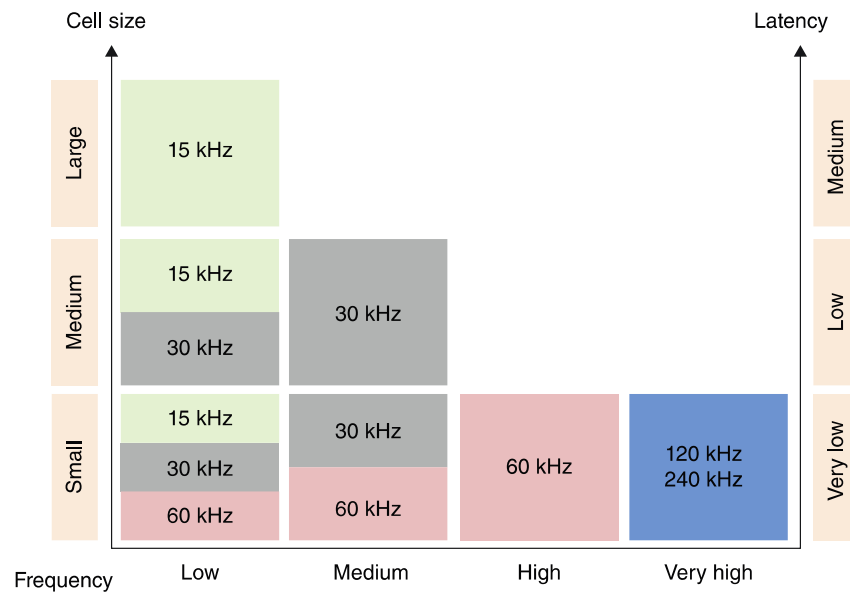


Figure 32.8 Typical use of different subcarrier spacings in a variety of application scenarios.

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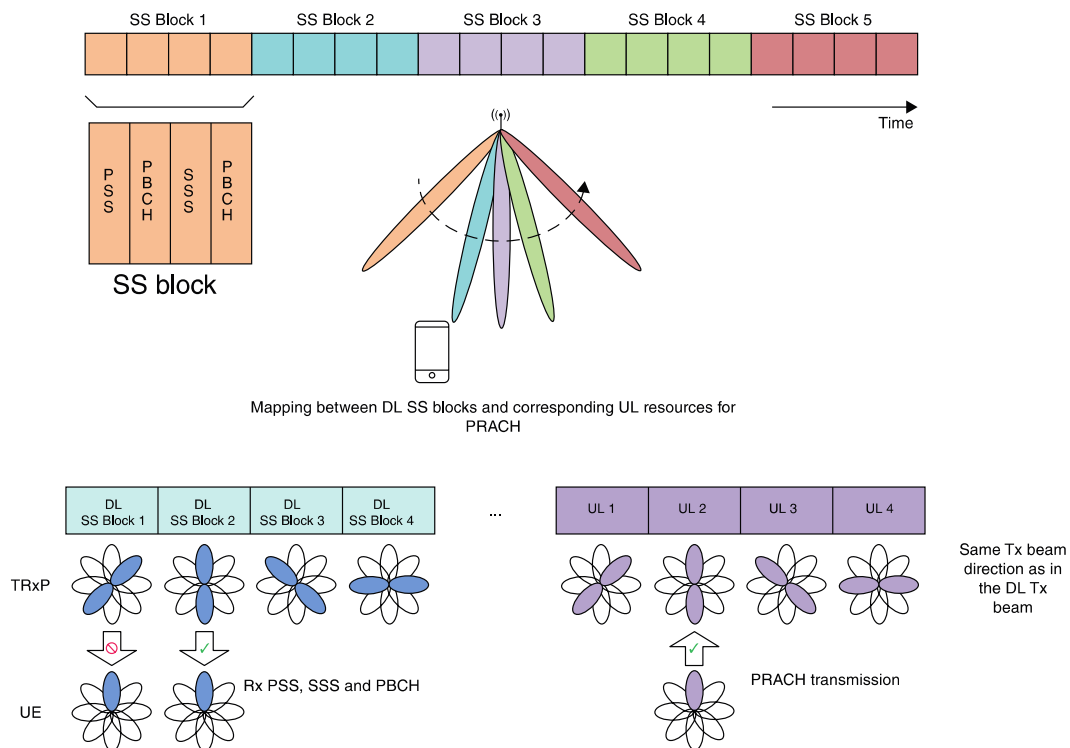


Figure 32.14 Establishment of beams at BS (TRxP) and UE. UE determines the best BS beam by monitoring different repetitions of the SS Block, which are transmitted into different directions. UE then transmits PRACH when the BS points the beam to it again, thus indicating to BS the best beam direction.

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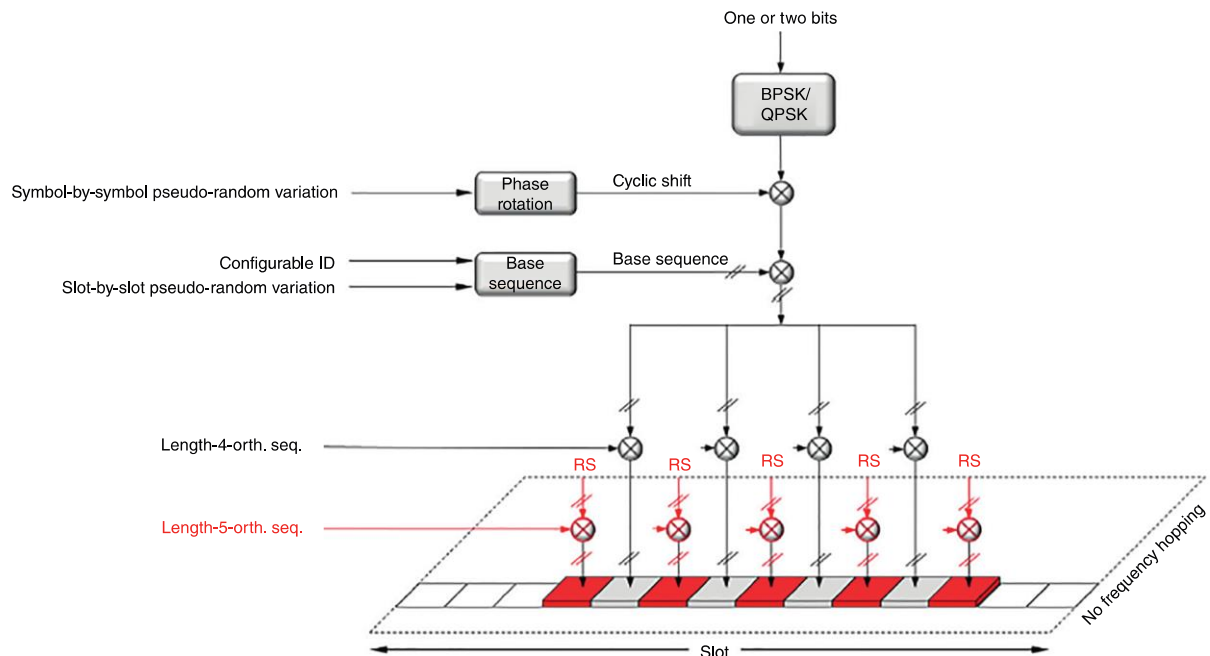


Figure 32.15 PUCCH format 1.

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Chapter 33

OPERATION IN U-NII BANDS – 802.11 CHANNEL PLAN (§15.407)

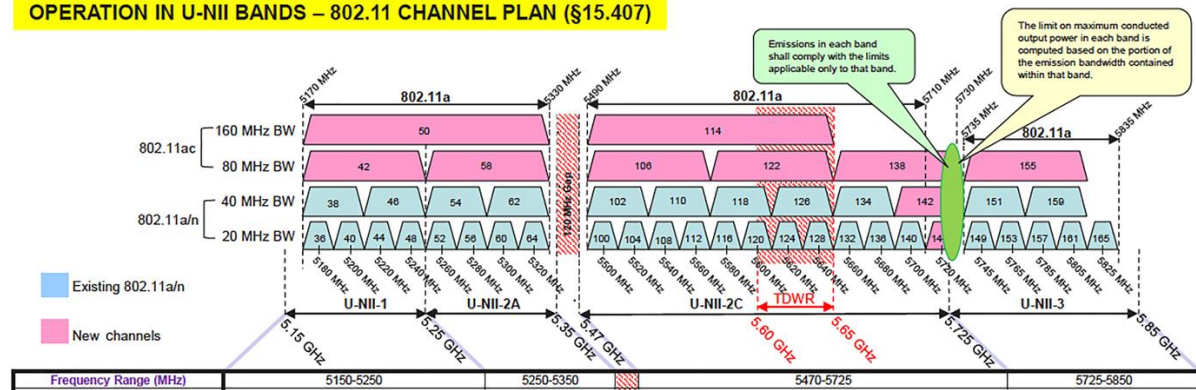


Figure 33.2 802.11 channel plan in the USA.

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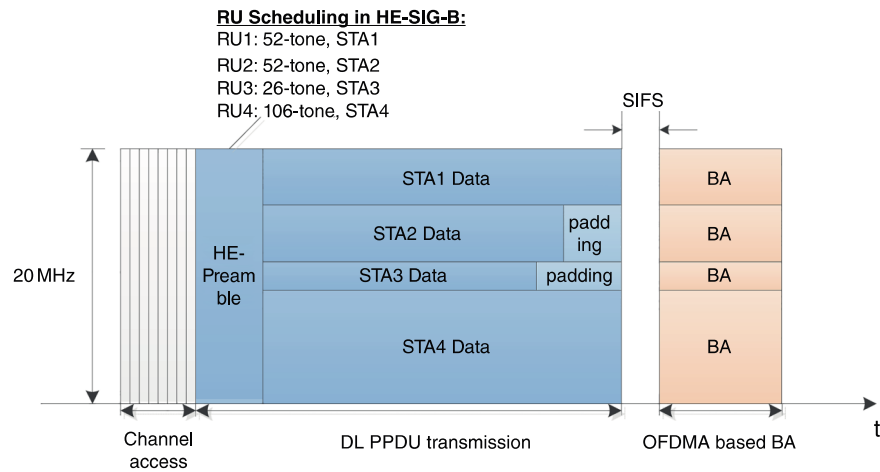


Figure 33.24 Example of a DLMU-MAC. Dark/mediumgrey: DL frame. Light grey: UL frame.

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Chapter 34

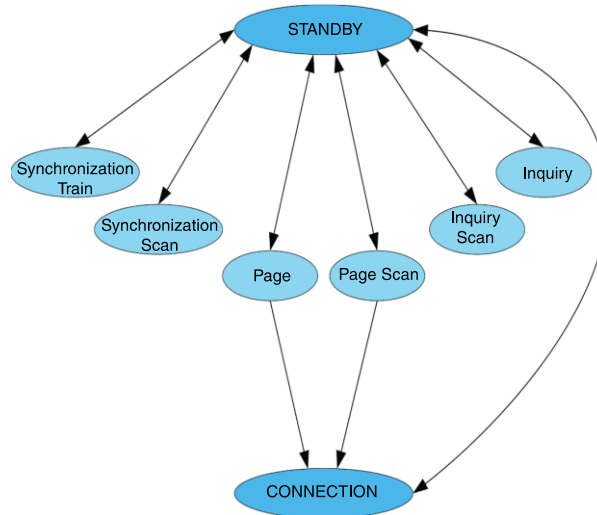


Figure 34.3 State diagram of link controller.

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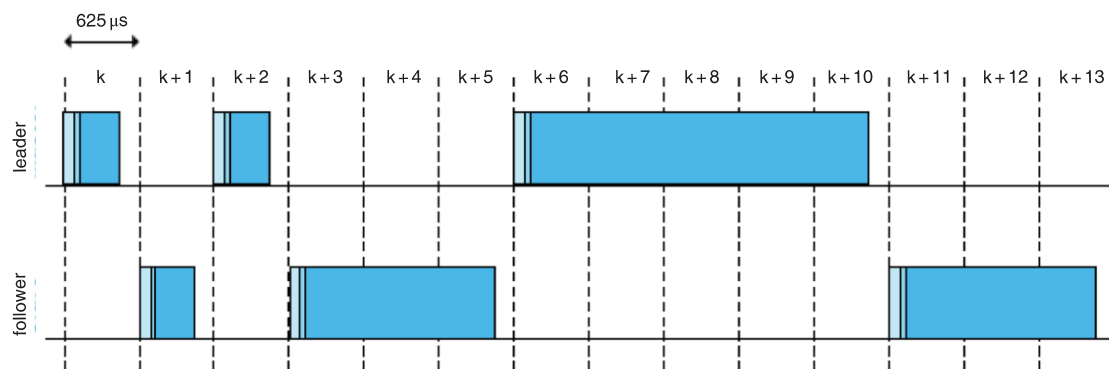


Figure 34.5 Slot structure in Bluetooth. Leaders transmit only packets starting in even timeslots, followers only in odd.

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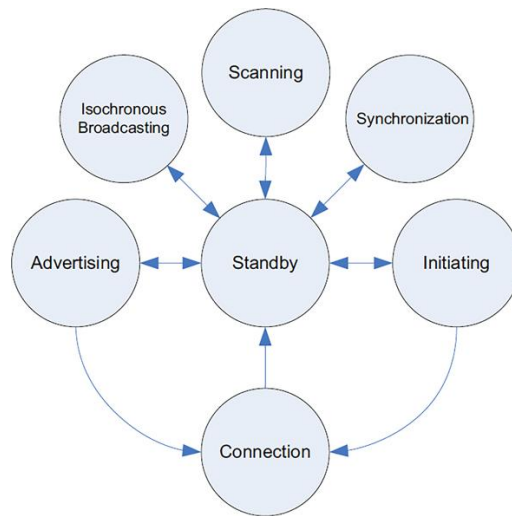


Figure 34.8 State diagram of the BLE link layer state machine.

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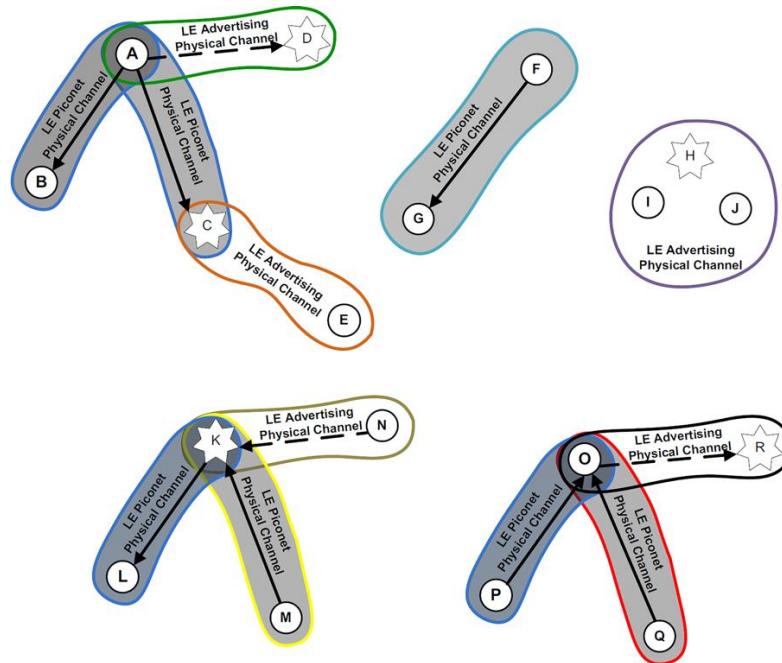
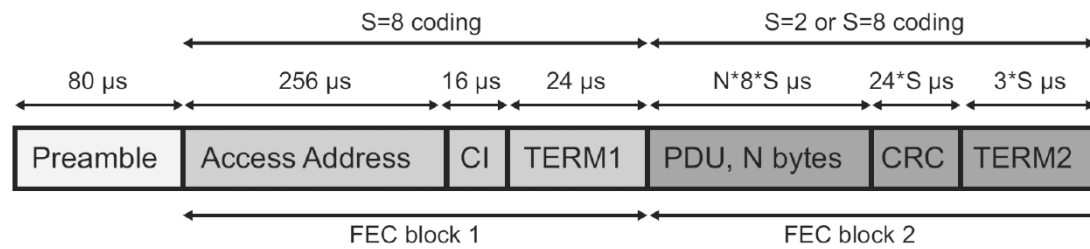


Figure 34.10 Solid arrows indicating connection leader to follower; dashed arrows, indicating a connection initiation, point from initiator to advertiser using a connectable advertising event; devices that are advertising are indicated using stars.

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Fields							
	Preamble	Access Address	CI	TERM1	PDU	CRC	TERM2
Number of Bits	Uncoded	32	2	3	16 – 2056	24	3
Duration when using S=8 coding (μs)	80	256	16	24	128 – 16448	192	24
Duration when using S=2 coding (μs)	80	256	16	24	32 – 4112	48	6

Figure 34.12 Packet format and field sizes and durations (in μs) for BLE coded PHY.

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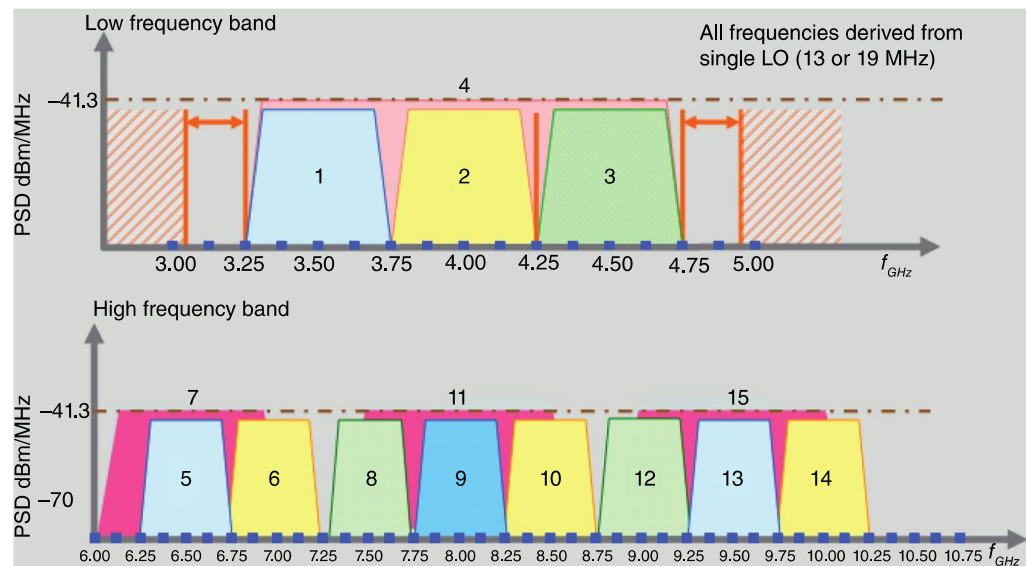


Figure 34.15 UWB bandplan for 802.15.4.

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